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Applied Data Science master thesis

Statistical Evaluation of Retrieval-Augmented Generation Systems Using Non-LLM Methods in the Insurance Domain

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Abstract

Large language models (LLMs) are increasingly used for complex question answering, but their tendency to produce ungrounded or factually incorrect outputs ("hallucinations") poses significant risks in regulated domains such as insurance. Retrieval-Augmented Generation (RAG) systems address this by grounding answers in retrieved policy clauses and structured documentation. However, evaluating their performance remains challenging due to transparency, reproducibility, and auditability requirements.

This thesis presents a statistical evaluation framework for RAG systems using only deterministic, non-LLM metrics. The framework applies 17 retrieval and 7 generation metrics to the Claim Assistant, a RAG-based tool for insurance policy comparison, using clause-level, expert-annotated datasets. The results show that the system retrieves and ranks relevant clauses more effectively for a.s.r. policies than for Aegon. However, generated outputs often achieve low BLEU and ROUGE scores despite being factually accurate, highlighting limitations of surface-level metrics.

By offering transparent and reproducible evaluation, this approach enables insurers to monitor and improve RAG systems without relying on black-box LLM scoring. The proposed evaluation framework combines clause-level ground truth datasets and deterministic retrieval and generation metrics to systematically evaluate system performance. It offers a reproducible approach for objective evaluation and continuous monitoring of retrieval and generation quality in real-world insurance applications.

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1. Introduction

Large language models (LLMs) such as GPT-3 [1] and GPT-4 [2] have become important tools in many industries. They can produce fluent text, summarize long documents, and answer complex questions. Because of their strong performance, LLMs are increasingly used for question answering systems, where users can ask natural-language queries and receive detailed responses. However, these models often face challenges with accuracy and trustworthiness. For example, LLMs can generate text that sounds correct but contains factual errors, a phenomenon often called hallucination [1][2][3][4].

To address this problem, researchers have developed Retrieval-Augmented Generation (RAG) methods. RAG combines LLM text generation with information retrieval techniques. The main idea is to retrieve relevant documents or passages that contain factual evidence and then use this information to guide the generation of the answer. This process helps reduce hallucinations and allows the system to stay up to date without retraining the model itself [1][3][5].

More specifically, a RAG system starts with a user query. It performs a semantic search in a document collection to find the most relevant content to that query. The documents retrieved are then added to the language model input to provide context for more accurate and grounded answers. In this way, retrieval mechanisms identify information relevant to the user question, and this evidence directly influences how the language model formulates its response [4][5]. As illustrated in Figure 1.1, it combines retrieval with generation.

[1] https://openai.com/index/gpt-3-apps/
[2] https://openai.com/index/gpt-4-research/

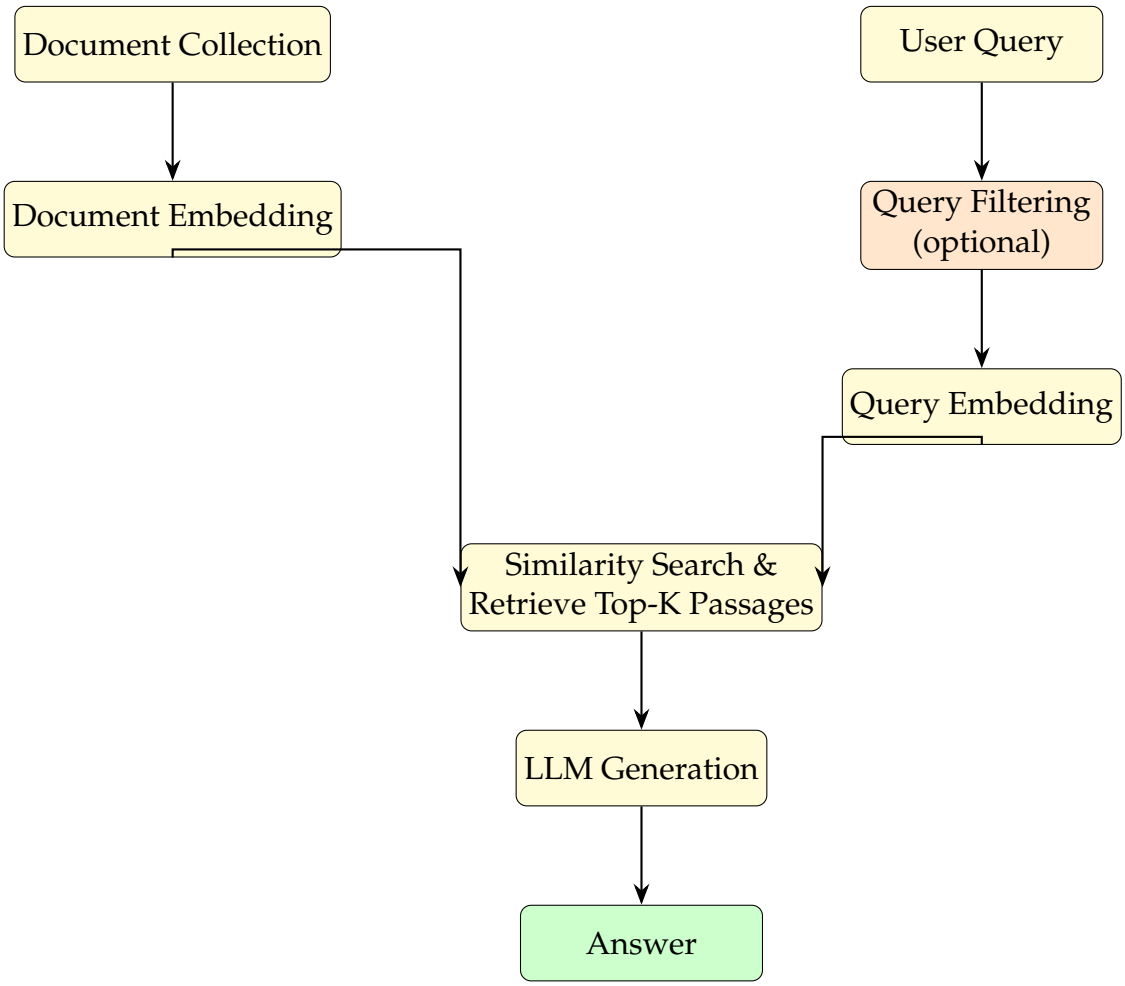


Figure 1.1: Retrieval-Augmented Generation (RAG) pipeline. The user query can optionally be filtered or reformulated and embedded, while the document collection is independently embedded and indexed. Similarity search compares query and document embeddings to retrieve the most relevant passages, which are then combined with the query to generate the final answer.

RAG systems have become important tools by combining retrieval mechanisms, which are responsible for finding supporting information, with generative models that produce coherent and context-aware answers. This dual process improves both precision and transparency compared to relying solely on LLMs [4].

As in many sectors, the popularity of RAG systems has been steadily rising in the insurance sector because they enable responses that are firmly rooted in authoritative policy documents and regulatory materials [6]. On the other hand, potential errors in automated outputs, such as improperly cited policy clauses or inaccurate statements, can lead to significant finan-

cial liabilities, legal penalties, and reputational harm [7]. Thus, to prevent these regulatory consequences, it is critical to employ robust evaluation methodologies for RAG systems that ensure transparency, reproducibility, and compliance with auditing standards.

To meet these demands, a wide range of evaluation methods have been proposed, including automatic metrics, human evaluations, and hybrid frameworks [7]. Among these approaches, and in light of the accomplishments of LLMs in a variety of domains, the notion of employing LLMs as evaluators, often termed 'LLM-as-a-Judge' [8], has gained considerable attention. In this approach, LLMs assess whether a generated response satisfies predefined rules, which may include using specific output formats (e.g., numerical scores or binary answers), or making decisions based on preferences as defined in evaluation prompts [9]. Although successfully implemented in several domains, these methods pose major practical difficulties in regulated sectors such as insurance. They often result in subjective evaluations due to sensitivity to prompts, demand high computational resources, and lack transparency for audit purposes [10]. Furthermore, the continuous evolution of LLMs can lead to variability in scoring outcomes, which affects reproducibility and accountability in enterprise settings.

These challenges have led to growing interest in non-LLM evaluation approaches. Deterministic, interpretable metrics, such as precision, recall and citation alignment, offer scalable, transparent, and audit-compliant alternatives. Rooted in the traditions of information retrieval (IR) and QA benchmarking, these metrics better align with the operational, regulatory, and risk-related constraints faced by insurance providers [11][12].

In this study, we evaluate the Claim Assistant system, an internal RAG tool designed to help users make informed insurance decisions. The system allows users to ask free-text questions about their coverage and receive structured, clause-level answers from the official documents of both Aegon and a.s.r. Given a user question and metadata (for example, product type, customer type, and coverage type), the system performs semantic search

with a FAISS [3]-based vector index and generates a side-by-side response using a GPT model deployed through Azure OpenAI [4]. The answer provides citations and article numbers from both insurers, highlighting key differences clearly. This allows users to assess their current coverage and explore alternative options under the same conditions. By automating this comparison process, the system eliminates the need for manually reading and interpreting long policy documents and empowers users to understand and potentially switch to the most suitable policy terms.

Despite its advantages, evaluating the Claim Assistant system poses practical challenges for a.s.r. As the system evolves, the insurer needs a reliable way to compare different RAG configurations, such as updated prompts, model versions, or retrieval strategies, while ensuring transparency, reproducibility, and auditability. In the absence of formal evaluation tools, the only available method has been asking end users which version they prefer, which is subjective, inconsistent, and time-consuming. However, since Claim Assistant grounds its responses in identifiable policy clauses, it offers a unique opportunity for objective evaluation using deterministic, article-level metrics. This leads to the central research question of the thesis: *How can we evaluate a domain-specific RAG system in a reproducible, transparent, and robust way using non-LLM metrics?* By addressing this question, the study aims to support a.s.r.'s goal of establishing a robust framework for monitoring and improving RAG-based policy comparison system.

Given the above challenges and considering the regulatory constraints specific to the insurance sector, our goal in this study is to use only deterministic and interpretable non-LLM metrics for evaluating the system's performance. Therefore, we adopted traditional IR and QA benchmarking metrics, such as precision, recall and similarity measures, instead of LLM-based evaluation methods. Our evaluation framework is fully decoupled from the live model and works on static outputs only. All evaluation metrics are computed externally to ensure consistency across experiments.

[3] `https://faiss.ai`

[4] `https://azure.microsoft.com/en-us/products/ai-services/openai-service`

To support this evaluation, we constructed two types of datasets: structured internal question sets aligned with policy metadata (Inboedel and Reis), and an additional test set of 9 real-world production questions. The latter were annotated manually by domain experts to produce gold-standard responses and citations. This setup allows us to test the system both in controlled and operationally realistic conditions.

Our contributions in this thesis are threefold:

- We process and enrich article-level and answer-level ground truth datasets across two insurance products, structuring them with standardized metadata for filtering and evaluation.
- We implement a set of deterministic, non-LLM evaluation metrics, covering retrieval correctness, ranking quality, faithfulness, and generation similarity.
- We apply the same evaluation suite to a manually labeled test set of 9 real-world production queries, enabling external validation of model performance under live usage scenarios.

The remainder of this thesis is structured as follows. Chapter 2 reviews relevant work on RAG architectures, evaluation methodologies, and domain-specific applications in the insurance sector. Chapter 3 first describes the Claim Assistant system pipeline, and then describes the construction and enrichment of evaluation datasets, including structured internal queries and expert-annotated test sets. Chapter 4 describes the technical implementation of the evaluation pipeline, including metric design. Chapter 5 presents experimental findings and performance comparisons across datasets and metric categories. Finally, Chapter 6 discusses limitations, practical implications, and directions for future work.

2. Literature Review

2.1 Why RAG needs measurement & What are the challenges?

Current studies [4][5] focus on the multi-faceted evaluation and comprehensive analysis of retrieval, generation, and the RAG system as a whole. However since each presents specific challenges, developing a comprehensive evaluation framework and benchmarks for RAG systems is difficult [7].

Moreover, in regulated industries such as insurance, where any incorrect clause or misinterpreted statute can lead to legal liability or financial loss, we therefore require rigorous, reproducible metrics for compliance, speed and cost control [7]. The following section will highlight some of these challenges.

- **Retrieval:** The retrieval process ensures that the most valid information is efficiently identified to support the generation component. Firstly in the indexing phase, documents are organized either through inverted indexes for sparse retrieval or dense vector embeddings for dense retrieval. During the search phase, these indexes are used to retrieve relevant documents in response to a user's query, often with optional re-ranking to improve the quality of results [13].

  However, there are a couple of challenges in evaluating this process, one major challenge is dynamic knowledge environments, which require evaluation metrics that can account for evolving data sources and time-sensitive shifts in relevance [14]. Furthermore, the nature of queries such as their ambiguity, user-specific phrasing, and temporal sensitivity makes it difficult to consistently measure precision, recall, and relevance of the retrieved documents, especially when considering

the accuracy of the data in a fast-paced environment [15].

- **Generation:** The generation component uses the retrieved content and the user's query to create coherent and contextually relevant responses through prompting and inference. The model dynamically interprets the context and evidence provided by the retrieval step, ensuring that the output is not only fluent and informative but also grounded in the available knowledge sources. This approach enables the system to deliver precise and insightful responses without the need to further fine-tune the model for specific tasks [3][5].

  For this process, the main challenges are to ensure that the answers are accurate and based only on the information retrieved, keeping the responses relevant to the question, and checking that the output is clear and well-written [5][15].

- **System as a whole:** Evaluating a RAG system as a whole is more complicated than simply checking its parts. It requires metrics that show how well retrieval and generation work together. It also needs to consider real-world issues like speed and how well the system can handle more users. Last but not least, it is important to look at how mistakes in retrieval affect the answers quality. Overall, the system should not be fully evaluated by looking at each part separately. It is essential to understand how everything fits together [7][16].

Having outlined the above challenges for retrieval, generation, and the system as a whole, the following section will delve into the literature on the deterministic metrics specifically designed for retrieval and generation components. For insurance providers, reliable metrics are essential to track whether the system meets business goals such as reducing manual effort in answering questions and improving customer satisfaction by providing correct information. Therefore, any evaluation framework must show not only if the retrieval and generation are working, but also whether they contribute to operational objectives.

2.2 Retrieval metrics

In the current literature [13][17], retrieval evaluation metrics try to balance traditional information retrieval (IR) approaches with RAG-specific adaptations. Building on this foundation, our evaluation framework adopts similar terminology while applying it to a RAG-based insurance domain. In this study, a query is a user question (e.g., *"Am I covered for cancellation due to illness?"*), and a document is a retrieved policy clause or article from the knowledge base (e.g., *"Article 2.5: Cancellations are covered under serious illness conditions."*). A document or article is relevant if it addresses the query based on the ground truth, that is, the set of gold articles manually associated with each query.

Accuracy measures the proportion of true results, both true positives (correctly retrieved relevant documents, TP) and true negatives (correctly excluded irrelevant documents, TN), relative to the total number of classifications. Here, false positives (FP) refer to irrelevant documents that are incorrectly retrieved, and false negatives (FN) are relevant documents that were not retrieved [13]. For a perfect model, the sum of true positives and true negatives equals the total number of classifications, resulting in an accuracy of 1.0.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Recall@k evaluates retrieval completeness. It is defined as the number of relevant items retrieved at position top k divided by the total number of relevant items available for the query. In a perfect retrieval system, all relevant documents (as defined by the ground truth and assumed to exist in the indexed knowledge base) would be present in the top k results, resulting in a recall@k of 1.0 and zero false negatives [17].

$$\text{Recall@k} = \frac{\text{Number of relevant documents in top } k}{\text{Total number of relevant documents}}$$

Precision@k assesses the precision of retrieval by measuring the proportion of relevant documents among the top k retrieved items. Mathematically, it is the number of relevant items in the top k results divided by k. In the ideal retrieval scenario, perfect model would return only relevant documents in the top k positions, achieving a precision@k of 1.0 and zero false positives [13][17].

$$\text{Precision@k} = \frac{\text{Number of relevant documents in top } k}{k}$$

As false positives decrease, precision increases; as false negatives decrease, recall increases. However, these metrics often exhibit an inverse relationship: increasing the classification threshold to reduce false positives can increase false negatives, and vice versa. This trade-off is well-documented in the literature, where improving one metric typically comes at the expense of the changing the other [13][17].

High values for both precision and recall are essential for building trustworthy retrieval systems. The **F1-score**, which is the harmonic mean of precision and recall, provides a single scalar value that balances both metrics. Since it penalizes large differences between precision and recall, it is particularly useful and a robust indicator of overall retrieval quality and also enables to optimize both metrics simultaneously [17].

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Mean Reciprocal Rank (MRR) is a widely used metric to evaluate the quality of the ranked results in information retrieval systems. It is calculated as the average of the reciprocal ranks of the first relevant document across a set of queries. Specifically, for each query, the reciprocal of the rank at which the first relevant document appears is computed (e.g., rank 1 yields a reciprocal rank of 1, rank 2 yields 0.5, and so on) [13].

In RAG systems, only the selected k retrieved documents are passed to the generation step, so it is common to calculate **MRR@k**. This means that the reciprocal rank is computed only if the relevant document appears within the first k positions. It is set to 1 when ranked first, or 0 if not found in the top k.

$$\text{MRR@k} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}$$

where rank_i is the position of the first relevant document within the top k retrieved results for the i-th query, and $|Q|$ is the total number of queries. MRR@k is particularly suitable for scenarios where the user is interested in the first relevant result among the top k candidates, such as question answering. However, it does not consider the presence of additional relevant results beyond the first one.

Mean Average Precision (MAP) provides a more comprehensive evaluation of ranking effectiveness by averaging precision values at each position where a relevant document occurs, across all queries. For each query, the precision is computed at every point in the ranked list where a relevant document is found, and these precision values are averaged to obtain the Average Precision (AP) for that query. MAP is then the mean of AP scores, ranging from 0 to 1, over all queries:

$$\text{AP} = \frac{1}{m} \sum_{k=1}^{n} \text{Precision}(k) \cdot \text{rel}(k)$$

$$\text{MAP} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \text{AP}_i$$

where m is the number of relevant documents for the query, $\text{Precision}(k)$ is the precision at rank k, and $\text{rel}(k)$ is an indicator function that is 1 if the document at rank k is relevant and 0 otherwise. MAP is especially useful when multiple relevant documents exist and when the user benefits from having them ranked as highly as possible, as it rewards systems that place all relevant documents near the top of the ranked list [13].

Coverage measures the percentage of queries for which the system fails to retrieve any ground truth (correct) documents. In other words, it quantifies the system's inability to find relevant information for certain queries, highlighting gaps in retrieval effectiveness. High coverage is desirable, as it indicates that the system is capable of retrieving relevant documents for the majority of user queries. It ranges from 0 to 1, where 0 means all queries had at least one retrieved document, and 1 means no queries were answered [4].

$$\text{Coverage} = \frac{\text{Queries with no match}}{|Q|}$$

Misleading Rate quantifies the proportion of retrieved content that does not align with the set of reference documents defined for the system. This metric ranges from 0 to 1, where 0 means all retrieved contents are within the reference set, and 1 means none of them are. A low misleading rate is especially important in domains like healthcare, law and insurance, where the accuracy of retrieved content directly impacts the trustworthiness of the

system [18].

$$\text{Misleading Rate} = \frac{\text{Number of misleading documents}}{\text{Total number of retrieved documents}} \times 100\%$$

Having discussed the main metrics for retrieval evaluation, the next section focuses on how the quality of generated responses is evaluated in RAG systems.

2.3 Generation-Quality metrics

Generation means producing an answer in natural language using the retrieved documents and the user question [4]. A diverse set of metrics has been proposed and adopted in the literature, each targeting different aspects of generation quality, ranging from exact lexical matches to deeper semantic and structural comparisons. This section reviews key metrics used in the evaluation of generated content, highlighting their strengths, limitations, and typical use cases [19].

Generation Accuracy (Exact Match) measures whether the generated output exactly matches the reference answer. This metric is straightforward but often overly strict, as it does not account for paraphrases or semantically equivalent responses. Its value is either 0 (no exact match) or 1 (exact match) [20].

ROUGE Recall-Oriented Understudy for Gisting Evaluation (ROUGE) computes n-gram overlaps between the generated and reference texts. ROUGE-N (unigrams, bigrams, etc.) and ROUGE-L (longest common subsequence) are widely used for summarization and text generation evaluation. ROUGE scores range from 0 to 1, where higher scores indicate greater similarity to the reference [21].

BLEU Bilingual Evaluation Understudy (BLEU) is a precision-based metric originally developed for machine translation. It compares the precision

of n-grams in the generated text compared to the reference texts, adjusting for the proper length and penalizing overly short translations through a brevity penalty. The BLEU score ranges from 0 to 1, where 1 indicates a perfect match with the reference translation. Although BLEU was first used for translation, it is also applied in question-answering and RAG evaluation, especially to measure how much surface-level overlap aligns with reference answers [22][23].

METEOR Metric for Evaluation of Translation with Explicit ORdering (METEOR) introduces additional features such as stemming, synonymy, and paraphrasing, providing a more flexible and semantically aware evaluation than BLEU. It is calculated using a generalized concept of unigram matching and includes penalties for fragmentation and word order. It also ranges from 0 to 1, with higher values indicating better alignment [24].

Cosine Similarity measures the similarity between vector representations (e.g., TF-IDF vectors or sentence embeddings) of the generated and reference texts. Because embeddings are often generated using LLMs, cosine similarity cannot be considered as a strictly non-LLM metric. [7]. However, in this study, it is treated as an exception due to its widespread usage in retrieval and generation evaluation. Cosine similarity values range from –1 (completely opposite vectors) through 0 (orthogonal vectors, 90 degrees to each other, no similarity) to 1 (identical vectors). This metric is sensitive to semantic relationships and is widely used in retrieval and generation tasks.

Jaccard Similarity computes the overlap between the sets of words in the generated and reference texts. It ranges from 0 (no overlap) to 1 (all words shared). It is a simple, set-based metric that quantifies lexical overlap, but does not account for word order or semantic nuance.

Edit Distance quantifies the minimum number of word-level or character-level operations (insertions, deletions, substitutions) required to transform the generated output into the reference. This metric and its variants are commonly used to measure the similarity between strings, with lower distances indicating greater similarity [25][26].

Although some of these metrics, like BLEU and METEOR, were originally

developed for translation tasks, they have been adapted and used in question-answering and RAG evaluation, especially to measure how much surface-level overlap aligns with reference answers [20][22][23][24].

In addition, when there are multiple correct answers to the same question, these metrics can be sensitive to wording variations. For example, BLEU and ROUGE might give low scores even if the generated text is factually correct but phrased differently. Therefore, it is often recommended to combine lexical metrics with embedding-based measures (such as cosine similarity) and ground truth to better capture the quality of diverse valid answers [21][23][27].

Despite progress in RAG evaluation, several critical gaps remain. For instance, the interdependence between retrieval precision and generation faithfulness remains underexplored in current studies, and there is no publicly available clause-level annotation set where each unit of retrieval corresponds to a numbered legal clause [4][7].

In summary, this literature review demonstrates that multi-level, deterministic evaluation is essential for RAG systems, especially in regulated domains. This study focuses exclusively on static, content-level metrics for retrieval and generation quality, and all analyses are performed offline on precollected model outputs. The following chapters describe how these insights inform the design and implementation of a reproducible evaluation framework designed for the insurance sector.

3. System Architecture and Evaluation Data

3.1 System Overview

The Claim Assistant is a RAG system designed to answer insurance-related queries by retrieving relevant clauses from official policy documents provided by two insurers, Aegon[1] and a.s.r.[2]. Given a user question and structured metadata (e.g., product type, customer type, and optional coverage), the system performs semantic search and answer generation in a multistep process. It first classifies the query, retrieves the top matching policy segments using a FAISS based vector index, and then generates a structured response using a GPT model deployed via Azure OpenAI. The response also provides citations and article numbers for both insurers, and a GPT-based comparison module highlights any differences in a clear format.

The pipeline consists of modular components deployed in Azure Machine Learning [3] (Azure ML), each responsible for a specific task: chunking, embedding, indexing, and inference. Policy documents are first split into semantically coherent chunks, and then are embedded using OpenAI's `text-embedding-3-large` [4] model, and indexed using FAISS. At inference time, the user query is embedded and semantically matched against this index. The top-ranked chunks are filtered based on metadata (e.g., insurance product and coverage type) and composed into a prompt, which is then sent to the GPT model for an answer generation.

Core components such as `VectorDB`, `VectorSearch`, and `GPTAzureOpenAI` coordinate retrieval and generation processes. While GPT inference runs through Azure OpenAI, all preprocessing and retrieval logic are executed

[1] `https://www.aegon.com/`
[2] `https://www.asrnl.com/`
[3] `https://azure.microsoft.com/en-us/products/machine-learning`
[4] `https://platform.openai.com/docs/models/text-embedding-3-large`

externally. Each inference is triggered by a front end request to an Azure ML endpoint, which returns the final structured response to the user. A full breakdown of the pipeline architecture is explained in Appendix A.1.

The next section outlines the preprocessing logic that transforms system outputs and other files into a unified evaluation dataset. With the system architecture in place, we now turn to the evaluation data, consisting of structured user questions, model responses, ground truth labels, and extracted policy content, that form the basis for retrieval and generation analysis.

3.2 Data Overview

This thesis evaluates the end-to-end performance of a RAG system using multiple structured data files. The system produces clause-level answers for three insurance products: contents insurance (*Inboedel*), continuous travel insurance (*Doorlopende Reisverzekering*, hereafter *Reis*), and home insurance (*Woonhuis*). However, annotated ground truth is available only for *Inboedel* and *Reis*.

The core dataset consists of 45 questions, 33 for *Reis* and 12 for *Inboedel*, along with system-generated responses, ground truth (GT) answers for both products, and extracted knowledge base (KB) articles. Table 3.1 presents summary statistics for the main data files, including row and column counts and the role of each file in the evaluation pipeline. These files are processed and merged using dedicated preprocessing scripts and form the foundation for subsequent metric computation and analysis.

Table 3.1: Summary statistics for files used in the evaluation pipeline

File Name (.xlsx)	Rows	Columns	Description
Groundtruth_inboedel	12	9	Original pivoted Inboedel GT file
Groundtruth_reis	33	9	Original pivoted Reis GT file
groundtruth_inboedel_enriched	96	8	Unpivoted metadata-enriched version
groundtruth_reis_enriched	132	8	Unpivoted metadata-enriched version
inboedel_questions_responses	96	5	Model responses for Inboedel
reis_questions_responses	132	5	Model responses for Reis
parsed_inboedel_responses	96	11	Flattened and cleaned response fields
parsed_reis_responses	132	11	Flattened and cleaned response fields
all_kbs_combined	833	6	Extracted knowledge base articles to check system results
merged_all	115	24	All datasets combined for evaluation

Throughout the evaluation, data preparation involves a series of transformation and enrichment steps. First, the system-generated responses are carefully parsed. Then, the original ground truth files are enhanced by normalizing their format and extracting relevant metadata, enabling alignment with the system outputs. Finally, article numbers from the knowledge base, constructed from segmented and indexed Dutch insurance policy documents, are extracted and organized to support justification-based evaluation.

All components are subsequently merged into a unified dataset that enables metric computation at the query level. These steps convert raw, heterogeneous sources into a consistent structure, allowing for fair and systematic evaluation of the RAG system. The next section outlines these preprocessing procedures in detail.

3.3 Description and Preparation of the Data

This section presents the datasets, preprocessing steps, and data structures used in evaluating the RAG system. The system is queried using a set of 45 questions, targeting only the *Inboedel* and *Reis* insurance products. Ground truth data files are enriched with standardized metadata to enable precise row-level alignment with the model responses, allowing for a reliable join operation during the merging process. Furthermore, a justification-based evaluation is made possible through the content extracted from the

segmented knowledge base.

The ultimate goal is to merge all components into a unified dataset that supports both retrieval and generation metric computation. Importantly, the system returns relevant articles and answers from both insurers (Aegon and a.s.r.) for each user question. To construct the merged dataset, one ground truth answer per query is extracted for each insurer, ensuring that every row contains responses and GT answers from both Aegon and a.s.r., along with all necessary metadata and evaluation fields.

The final dataset includes columns such as the user question, product, coverage type (`dekking`), customer type (`type_klant`), policy version (`polis_versie`), parsed answers and citations from both Aegon and a.s.r., ground truth article numbers and answers, retrieved article numbers, model comparison summaries, and ground truth source names. An example row is presented in Table 3.2 to illustrate the structure of this merged dataset, with the original Dutch version provided in Appendix Table B.1.

Table 3.2: Example Row from the Final Merged Evaluation Dataset

Column	Value
question	I have a concussion, I want to cancel, is this covered?
product	Reis (Continuous travel insurance)
dekking (coverage type)	Basic
type_klant (client type)	Advisor
polis_versie (policy)	
comparison_response_-antwoord	Both insurers recognize concussion as a possible reason for cancellation under serious illness.
aegon_system_answer	Yes, a concussion is considered a serious illness, which is a valid reason for cancellation under the insurance policy.
asr_system_answer	yes
top_3_aegon	['2.5', '2.6', '2.2']
top_3_asr	['3.10', '4.4', '3.1']
system_article_ids_aegon	['2.5.1']
system_article_ids_asr	['3.10']
gt_article_ids_aegon	['2.5.1']
gt_article_ids_asr	['3.10']
aegon_gold_answer	yes, in point 1 it says "serious illness" in chapter 6 the glossary states what we mean by serious illness, concussion is not mentioned here but we have decided to no longer follow the list
asr_gold_answer	yes
gt_source_aegon	aegon_reis
gt_source_asr	asr_vp_dr_2024
kb_source_aegon	aegon_reis
kb_source_asr	asr_vp_dr_2024
sub_mentions_in_content	[]
mentions_other_articles	['3.1']

3.3.1 System Responses and Parsing

The evaluation process relies on accurately aligning system outputs with ground truth data. We begin by examining how the model response files are structured and parsed for this purpose. These output files are collected separately for the Inboedel and Reis question sets. Each structured file contains the original user question, relevant metadata fields such as `product`, `dekking`, `type_klant`, and, where applicable, `polis_versie`, as well as a single `response` column.

Each model response is stored as a nested dictionary under the `response` column. This field includes the generated answers, article numbers, and supporting quotes from both Aegon and a.s.r., along with a textual comparison and optional classification warnings.

To prepare these files for evaluation, the `response` field is parsed into explicit columns, while retaining the surrounding metadata. Additionally, top-3 retrieval results for each insurer are appended from separate output files.

After parsing and appending new columns, the final system response dataset consists of one row per user question, with side-by-side outputs from both insurers. This flattened format supports direct comparison and metric evaluation. The resulting column structure is summarized in Table 3.3.

Table 3.3: Final Column Structure of Parsed System Response File

Column Name	Description
`question`	Original user question
`product`	Product type (e.g., Inboedel, Reis)
`dekking`	Coverage type (e.g., basis, allrisk)
`type_klant`	Customer type (e.g., direct, via advisor)
`polis_versie`	Policy version label
`aegon_response_Antwoord`	Answer generated by the model for Aegon
`aegon_response_Citaat`	Supporting quote from Aegon policy
`aegon_response_artikel_nummer`	Article number cited by Aegon response
`asr_response_Antwoord`	Answer generated by the model for a.s.r.
`asr_response_Citaat`	Supporting quote from a.s.r. policy
`asr_response_artikel_nummer`	Article number cited by a.s.r. response
`comparision_response_Antwoord`	Textual comparison between both insurer responses
`top_3_retrieved_aegon`	Top-3 article numbers retrieved for Aegon
`top_3_retrieved_asr`	Top-3 article numbers retrieved for a.s.r.

In the next subsection, we describe the structure of the ground truth answers and explain how metadata fields such as `product`, `polis_versie`, `dekking`, and `type_klant` enable alignment between the system outputs and the corresponding gold articles and answers.

3.3.2 Ground Truth Data Structure and Enrichment

Like system output files, the original ground truth data is also provided separately for *Reis* and *Inboedel*. Each dataset contains manually annotated

answers for various insurer–policy combinations, structured in a pivot format where each policy variant appears as a separate column. While the pivoted format preserves the logical structure of the underlying policy documents, it is not directly compatible with the row-level alignment required for evaluation.

For example, the Inboedel product includes eight distinct policy variants, covering two `dekking` types (e.g., `basis`, `allrisk`) for each of four insurer–policy combinations. Table 3.4 shows the original layout, where each answer is paired with an article number under a separate column heading.

Table 3.4: Original Pivoted Column Structure for Inboedel Ground Truth

	a.s.r. IKZ		a.s.r. advies		Aegon oud 3038		Aegon nieuw 3041	
Coverage	basis	allrisk	basis	topdekking	basis	allrisk	basis	royaal
Question	art.no.	answer	art.no.	answer	art.no.	answer	art.no.	answer
...	...	...	...	...	...	...	...	...

A similar pivot structure is used for the Reis dataset. Table 3.5 illustrates how each column pair represents a policy variant (e.g., "Aegon basis"), with the article number and corresponding answer in separate columns.

Table 3.5: Original Pivoted Column Structure for Reis Ground Truth

	Aegon basis		Aegon allrisk		a.s.r. VP DR 2024		Ik kies zelf (DR 2018)	
	article	answer	article	answer	article	answer	article	answer
Section	...	...	...	...	...	...	...	...
Question	...	...	...	...	...	...	...	...

As with Inboedel, this pivoted layout has to be transformed into a row-wise format before evaluation. To achieve this, each article–answer pair is unpivoted into a separate row, and explicit metadata fields are added to support alignment with system outputs:

- `product`,
- `dekking` (coverage type),
- `type_klant` (customer type),
- `polis_versie` (policy version), and

- `gt_source` (e.g., `asr_vp_dr_2024`)

This enrichment step is essential because the system output files already contain metadata fields, and accurate alignment depends on matching each question to the correct ground truth answer based on those fields. Without metadata enrichment, it is not possible to determine which of the eight Inboedel or four Reis variants should be used for evaluation.

The enriched ground truth data thus serves as a structured and queryable source of correct answers, organized by insurer and metadata combination. This enables the merging of parsed system responses with their corresponding ground truth answers, article numbers, and source identifiers for both Aegon and a.s.r., laying the foundation for computing retrieval and generation metrics.

To illustrate how the pivoted structure is transformed into enriched clause-level data, Table 3.6 shows a sample row from the Inboedel dataset. This example includes the original user question, the corresponding gold-standard answer, and the metadata fields necessary for matching. The original Dutch version of this row is provided in Table B.2.

Table 3.6: Sample Enriched Row – Inboedel (English Translation)

Question	Art.	Answer	GT Source	Product	Customer	Coverage
A ball broke my window. Is this covered?	3.3	No, unless glass is additionally insured.	a.s.r._ikz_basis	Inboedel	Direct	Basis

To demonstrate the structure of the enriched Reis dataset, Table 3.7 presents a representative example. It includes a user question, the corresponding gold-standard answer, and all metadata fields required for evaluation. This particular example is derived from the Aegon basis policy variant. The original Dutch version of this row is available in Table B.3.

Table 3.7: Sample Enriched Row – Reis (English Translation)

Question	Answer	Article	GT Source	Product	Customer	Coverage
I have a concussion and want to cancel my trip. Is this covered?	Yes, in point 1 "serious illness" is listed. While concussions are not mentioned in Chapter 6, it was decided to no longer follow that list.	2.5.1	aegon_basis	Continuous Travel Insurance	(blank)	Basis

3.3.3 Knowledge Base Extraction

The knowledge base (KB) used in this thesis is not manually constructed but is automatically produced during the model deployment. It consists of preprocessed Dutch insurance policy documents, segmented and embedded as part of the RAG pipeline. The KB is provided in JSONL format and covers multiple product versions from both insurers, including:

- Aegon *Inboedel* (Old and New),
- a.s.r. *Inboedel* (Standard and IKZ),
- Aegon *Reis*,
- a.s.r. *Reis* (VP DR 2024 and IKZ DR 2018).

Although the system already returns the top-k articles for each query, this top-k list alone does not fully reveal whether the retrieved content actually supports or justifies the answer. For example, the system might return article `1.14.4` as its top result, while the gold article is `1.18`. If `1.14.4` explicitly refers to `1.18`, the model's output may still be valid and justified, even if it is not an exact match.

To capture such indirect relationships between articles, a separate extraction step is performed. A script parses each KB file to collect the `article_number` and `content`, and recursively searches for any references to sub-articles or other articles mentioned in the text. This results in two additional columns:

- `sub_mentions_in_content`: listing sub-articles such as `1.14.1`–`1.14.4`,
- `mentions_other_articles`: capturing direct references to other articles, e.g., `1.18`.

Table 3.8: Knowledge Base Extraction Example

Field	Content
article_number	1.14
content	`1.14 When does the insurance start and end?` `1.14.1 Starts on the policy date. Earlier events not covered.` `1.14.2 Cancel anytime after first year. Refund applies. Extra cover can be removed or downgraded.` `1.14.3 We cancel immediately for fraud, non-disclosure, payment or cooperation issues.` `1.14.4 We cancel with notice for non-payment (see 1.18), moving abroad, renewal, or claims.`
sub_mentions_-in_content	['1.14.1', '1.14.2', '1.14.3', '1.14.4']
mentions_other_-articles	['1.14.1', '1.14.2', '1.14.3', '1.14.4', '1.18']
kb_source	aegon_reis
kb_source_group	aegon

The extracted and enriched knowledge base is saved as a separate data file, consisting of over 830 rows in 6 columns. This data plays a crucial role in later stages of evaluation, especially when analyzing whether a system-selected article can be considered justified through indirect links. These relationships are not observable from the top-k results alone and are revisited in the evaluation methodology when discussing faithfulness and justification-based metrics.

3.3.4 Final Merging and Evaluation Dataset

As outlined earlier, all processed components, system outputs, ground truth answers, and extracted knowledge base entries are merged into a single evaluation-ready file. The final combined dataset contains 115 rows and 36 fields in total. Each row corresponds to a unique user query and includes all necessary fields, such as original metadata, parsed system responses, top-k retrieval results, ground truth labels, and KB-based justification fields. An example of its structure is provided in Table 3.2 in Section 3.3. This dataset serves as the unified foundation for comprehensive, query-level computation of all retrieval and generation metrics discussed in the next chapter.

3.3.5 Test Dataset: Real-World Queries

To complement the structured evaluation set, a separate test set containing 9 real-world queries submitted to the production version of the Claim Assistant system is also included. These user-submitted questions are manually reviewed by domain experts, who annotate the correct policy clauses from the official insurance documents.

Each entry in this test set includes:

- The original user question
- The system-generated answers and article references
- Expert-curated gold answers and citations for both insurers

This real-world test set serves as an external benchmark for validating model performance under actual deployment conditions. The same retrieval and generation metrics are applied to this dataset, enabling direct comparison with internally generated results. A detailed analysis of these real-world examples is provided in the next chapters.

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4. Methodology

The performance of the Claim Assistant RAG system is evaluated using a combination of retrieval-specific and generation-specific metrics. The evaluation process does not involve modifying the model itself but relies solely on comparing system outputs to a manually defined ground truth. The following sections describe the methodology used to carry out the evaluation process.

4.1 Evaluation Data Preparation

Before calculating retrieval and generation metrics, we added several new columns to the original dataset. These additional fields, created using Python preprocessing functions, were appended alongside the original columns and are included in all evaluation output. The full implementation of the functions used to generate these fields is provided in Appendix C. They help interpret the results more clearly and include the following:

- **gt_quality_issue**: This field shows if the ground truth data for a question was incomplete or unclear. For example, sometimes there was no article number, only a page reference. When this is TRUE, it means that the evaluation of this query should be interpreted carefully.
- **match_type**: This field describes how the retrieved article compares to the gold article number. It can be `exact` (the same article number), `prefix` (the retrieved article is an article that includes the gold article as a sub-article), or `no_match` (no numbering relation).
- **justified_by_kb**: This field is TRUE if the retrieved article is considered justified by the knowledge base. To check this, we verified whether the retrieved article text contained references to the gold article. This included two checks: (1) if the retrieved article listed the gold article

as a sub-article (`sub_mentions_in_content`), or (2) if the text explicitly mentioned the gold article (`mentions_other_articles`). This makes it possible to accept indirect references as valid justification even without an exact match.

- **kb_source**: This shows which knowledge base version was used for the justification check (for example, `aegon_reis` or `asr_vp_dr_2024`).

These fields were created before any metric calculations. They make it possible to explain why an article was counted as correct or justified. They also help to check the reliability and faithfulness of the evaluation.

4.2 Retrieval Evaluation Method

The retrieval performance of the Claim Assistant system is evaluated by comparing the article numbers returned by the deployed model against manually defined ground-truth identifiers for each query. These ground truth fields, `gt_article_ids_asr` and `gt_article_ids_aegon`, are derived from insurance policies and contain the correct article that should be retrieved for a given question.

To ensure that comparisons are contextually valid, each ground truth row is paired with structured metadata, including product types, such as Inboedel and Reis, policy version such as 2018 or 2024, and specific coverage types such as all-risk or basic. These metadata fields are also present in the system outputs. Before performing any comparison, the evaluation script filters the ground truth to retain only those rows whose metadata match the query context of the system output. In this way, article comparisons are only made within the same policy variant and coverage configuration, avoiding mismatches due to irrelevant documents.

The system's own output fields, `system_article_ids_asr` and `system_article_ids_aegon`, contain the article numbers selected by the model for each query. These are then compared against the metadata-aligned ground truth articles using a dedicated `classify_match()` function. This function categorizes each system prediction as either an `exact` match (full match), a

`prefix` match (partial or hierarchical match), or `no_match`.

To further evaluate retrieval quality, the evaluation process also verifies whether the predicted article logically justifies or supports the correct one. This is handled by the `is_justified_by_kb()` function, which scans the content of the retrieved article within the relevant knowledge base source (e.g., `kb_source_asr`, `kb_source_aegon`) and checks whether it includes, references, or semantically explains the correct gold article. These results are recorded in fields such as `justified_by_kb_asr`, allowing a more nuanced evaluation of whether a system response, even if not an exact match, can still be considered acceptable or misleading.

As already mentioned, all evaluation metrics are computed externally, without querying the vector index or modifying the system's inference behavior. Metric computation is performed at two levels:

- **Per-query level (row-level):** For each individual query in the evaluation dataset, a dedicated function, `compute_all_retrieval_metrics_dataframe()`, computes detailed retrieval outcomes such as match type (e.g., exact, prefix, or no match), justification status, and top-k retrieval accuracy. These per-row metrics allow fine-grained inspection of the system's behavior on a query-by-query basis.
- **Aggregate level (system-level):** After computing metrics for all individual queries, the results are aggregated across the entire dataset using `compute_row_metrics_for_source()`. This produces overall performance statistics such as Retrieval Accuracy, Recall@k, MRR, and Misleading Rate, enabling system-wide evaluation and comparative analysis between insurers.

This dual level evaluation design ensures that both granular query-specific behavior and holistic system performance can be analyzed in a consistent manner. The corresponding functions are listed in Appendix C.

The retrieval evaluation covers a total of 17 distinct metrics, which are grouped into four categories: accuracy, ranking quality, coverage, and faithfulness. These metrics are computed both at the system level (aggregated

across the full dataset) and at the per-query level (row-by-row). Each metric is calculated independently for the two insurers (a.s.r. and Aegon) to enable fair and comparable analysis. These metrics follow the formal definitions and equations presented in Chapter 2, specifically in Section 2.2.

1. Final Answer Accuracy Metrics:

- **Retrieval Accuracy (Exact):** Percentage of queries where the system returns an article that exactly matches a ground truth article ID.
- **Extended Accuracy (Exact + Prefix):** Also counts prefix-based matches, allowing for hierarchical article structures.

The Retrieval Accuracy and Extended Accuracy metrics are calculated by parsing the article number explicitly cited in the final answer generated by GPT. In other words, these metrics evaluate whether the article selected in the generation step matches the ground truth reference. By contrast, the following ranking metrics are computed using the top-k retrieval candidates produced by the system's semantic search step before any GPT generation takes place. This means that the article IDs considered in ranking metrics may differ from the final selected citation. For example, the retrieval component may return a list such as [2.5, 3.1, 2.2], but GPT may select a more specific sub-article like 2.5.1 when generating the answer. Therefore, selection-based and ranking-based metrics are not necessarily aligned.

2. Ranking Quality Metrics:

- **Accuracy@1:** Proportion of queries where a relevant article appears at the top-1 position. It is computed as a binary value per query: 1 if the top-1 retrieved article exactly matches the ground truth, 0 otherwise.
- **Recall@1:** Proportion of queries where the top-1 retrieved article is relevant based on the ground truth.
- **Precision@1:** Proportion of top-1 retrieved articles that are relevant.
- **Accuracy@3:** Proportion of queries where at least one of the top-3 retrieved articles exactly matches the ground truth.
- **Recall@3:** Proportion of queries where at least one of the top-3 retrieved

articles is relevant based on the ground truth.

- **Precision@3:** Proportion of the top-3 retrieved articles that are relevant.
- **F1@3:** The harmonic mean of Precision@3 and Recall@3. It reflects the balance between the number of relevant articles retrieved and the number of retrieved articles that are actually relevant within the top-3.
- **Mean Reciprocal Rank (MRR):** Average of the reciprocal ranks of the first relevant article retrieved for each query.
- **Mean Average Precision (MAP):** Average of the precision values at the ranks where relevant articles appear, calculated across all queries.
- **Coverage (No Match):** Proportion of queries for which no relevant article was retrieved.
- **Justification Rate (All Matches):** Proportion of queries for which all retrieved articles are justified via the knowledge base (i.e., it contains or cites the gold article).

4. Faithfulness and Misleading Metrics:

- **False Positive Rate (Exact):** Among exact matches, the proportion not justified by the knowledge base.
- **False Positive Rate (Extended):** Among exact or prefix matches, the proportion not justified.
- **Misleading Rate (Exact Not Justified):** Cases with exact matches that are unsupported.
- **Extended Misleading Rate (Exact or Prefix Not Justified):** Cases with exact or prefix matches that are unsupported.
- **Misleading Coverage (No Match & Not Justified):** Cases where no relevant article was retrieved and none of the retrieved articles are justified.

These 17 metrics are computed both *per query* using the `compute_row_metrics_for_source()` function, and *system-wide* using `compute_metrics_for_source()`, enabling both granular and aggregated performance insights.

More importantly, the evaluation modules analyze previously generated model outputs, specifically the retrieved article numbers, against the corresponding ground truth. This ensures that evaluation remains reproducible and does not interfere with model inference behavior.

4.3 Generation Evaluation Method

The quality of generated answers is evaluated by comparing the system-generated response (`Antwoord`) to a predefined ground truth answer (`gold_answer` for both Aegon and a.s.r.. The evaluation focuses on semantic similarity and surface-level lexical overlap. For each query, the following metrics are computed via the `compute_generation_metrics()` function:

Embedding-Based Metric

- **Cosine Similarity:** Computes vector similarity between embedded representations of the system and gold answers. For this purpose `text-embedding-3-large` is used.

String-Based Metrics

- **Jaccard Similarity:** Word overlap ratio between system and gold answers.
- **Edit Distance:** Edit distance between the two strings, representing the number of insertions, deletions, or substitutions required.

Lexical Metrics

- **BLEU Score:** Measures n-gram overlap with smoothing; useful for sentence-level comparison.
- **ROUGE Scores (ROUGE-1, ROUGE-2, ROUGE-L):** Evaluate unigram, bigram, and longest common subsequence overlap.

All metrics are stored as additional columns in the output file, allowing for in-depth analysis at both system and per-query levels.

4.4 Manual Evaluation on Real-World Queries

To complement the primary evaluation, we also apply the same metric framework to a manually labeled test set consisting of 9 real-world user queries submitted through the live system. For each question, domain experts identified the correct answer and citation from the original policy documents. These expert annotations are structured similarly to the internal ground truth format and allowed for consistent evaluation using the existing retrieval and generation metric scripts. This additional test enables validation of the system's robustness and accuracy under realistic usage conditions.

5. Results

5.1 Overview of the Results

This chapter presents the evaluation results of the Claim Assistant RAG system. The analyses are carried out using the deterministic framework described in Chapter 4. All results are reported separately for the merged dataset of 45 queries and the expert-labeled test set of 9 real-world user queries.

The evaluation covers both retrieval and generation performance. To make the evaluation process more transparent, we begin by introducing an example query that demonstrates how retrieved articles and generated responses are evaluated and scored. This example helps illustrate the logic behind each metric and clarifies how exact matches, prefix matches, and justification checks influence the final evaluation results.

After the example analysis, we present detailed retrieval results for the full merged dataset and then continue with an analysis of generation performance.

Finally, we apply the same evaluation procedure to the separate test dataset of 9 queries to examine whether similar patterns hold in real-world examples. The chapter concludes by summarizing the key findings, emphasizing the impact of retrieval accuracy on generation performance, and discussing how strict matching criteria and justification checks together provide a more complete evaluation of the Claim Assistant system.

5.1.1 Example Query Analysis

To make the evaluation outputs easier to understand, we first examine a single example query. Table 5.1 displays the relevant data for this query as produced by the system. Fields such as `match_type`, `justified_by_kb`,

and `gt_quality_issue` are created during evaluation data preparation (see Section 4.1). These fields indicate how each retrieved article is evaluated in relation to the ground truth.

Table 5.1: Input fields and metadata for the example query

Field	**Value**
Question	I have a concussion, I want to cancel, is this covered?
Product	Continuous travel insurance
Coverage	Basic
Type_klant	Advisor
Aegon Gold Answer	Yes, in point 1 it says "serious illness"...
a.s.r. Gold Answer	Yes
Aegon System Answer	Yes, a concussion is considered a serious illness...
a.s.r. System Answer	Yes, if you become seriously ill...
Top-3 Aegon	2.5, 2.6, 2.2
Top-3 a.s.r.	3.10, 4.4, 3.1
System Article ID Aegon	2.5
System Article ID a.s.r.	3.10
Gold Article ID Aegon	2.5.1
Gold Article ID a.s.r.	3.10
GT Quality Issue Aegon	FALSE
GT Quality Issue a.s.r.	FALSE
KB Source Aegon	aegon_reis
KB Source a.s.r.	asr_vp_dr_2024
Match Type Aegon	prefix
Match Type a.s.r.	exact
Justified by KB Aegon	TRUE
Justified by KB a.s.r.	TRUE

The following Table 5.2 presents a detailed view of how the retrieval choices of the system for this example query are reflected in the retrieval metrics. It helps illustrate how different match types and ranking positions affect the numerical scores.

Table 5.2: Retrieval Metrics for the Example Query. *Arrows indicate whether higher or lower values are preferable: ↑ means higher is better; ↓ means lower is better.*

Metric	**Aegon**	**a.s.r.**
Retrieval Accuracy (Exact) ↑	0	1
Extended Accuracy (Exact + Prefix) ↑	1	1
Recall@1 ↑	0	1
Precision@1 ↑	0	1
Accuracy@3 ↑	0	1
Recall@3 ↑	0	1
Precision@3 ↑	0	0.33
Mean Reciprocal Rank (MRR) ↑	0	1
Mean Average Precision (MAP) ↑	0	1
Coverage (No Match) ↓	0	0
Misleading Rate (Exact Not Justified) ↓	0	0
Extended Misleading Rate ↓	0	0
False Positive Rate (Exact) ↓	0	0
False Positive Rate (Extended) ↓	0	0
Justification Rate (All Matches) ↑	1	1
Misleading Coverage ↓	0	0

In this example, the system retrieves article `2.5` instead of the gold article `2.5.1`. Because `2.5` is an article that includes `2.5.1` as a sub-article, it is marked as a prefix match. However, metrics that require an exact match do not count this retrieval as correct. As a result, Retrieval Accuracy (Exact), Recall@1, Accuracy@3, Mean Reciprocal Rank (MRR), and Mean Average Precision (MAP) are all zero. This happened even though the retrieved article contains relevant information. At the same time, Extended Accuracy and Justification Rate are both one, because the system is allowed to count prefix matches and references in the knowledge base as valid. For a.s.r., the system retrieves the exact correct article, so all ranking metrics are one.

This example shows how using articles and sub-articles affects the evaluation results. Even when the retrieved content is relevant, strict metrics only count exact matches as correct.

Table 5.3: Generation Metrics for the Example Query. *Arrows indicate whether higher or lower values are preferable: ↑ means higher is better; ↓ means lower is better.*

Metric	**Aegon**	**a.s.r.**
Cosine Similarity ↑	0.5657	0.1582
Jaccard Similarity ↑	0.1429	0
Edit Distance ↓	149	149
BLEU Score ↑	0.0228	0
ROUGE-1 ↑	0.2083	0
ROUGE-2 ↑	0.0385	0
ROUGE-L ↑	0.1250	0

Table 5.3 shows the generation metrics for the same example query. In this case, the Claim Assistant system achieves substantially higher similarity scores for Aegon answer across all metrics. For example, Cosine Similarity is 0.5657 for Aegon and only 0.1582 for a.s.r., while Jaccard similarity reaches 0.1429 for Aegon but remains zero for a.s.r. BLEU and ROUGE scores also follow this pattern, with Aegon side showing small but positive values, whereas the a.s.r. side scores zero on most metrics.

This difference is primarily due to the length and wording of the gold reference answers. For the Aegon dataset, the gold answer includes a longer sentence describing "serious illness," which overlaps with the generated output. In contrast, the gold answer from the a.s.r. dataset consists of a single word ("Yes"), while the generated output is a longer sentence. This mismatch in length and vocabulary results in very low overlap scores for a.s.r., even though the generated answer is semantically correct.

Interestingly, Edit Distance is identical (149) for both sides, reflecting that a similar amount of editing is required to fully align each system-generated answer with its respective gold answers. This example highlights how short gold answers and longer generated sentences can strongly impact lexical similarity metrics, even when the overall meaning is similar.

After demonstrating how the evaluation metrics apply to an individual query, the next sections present retrieval results across the full dataset. This provides a broader perspective on how the system performs in general.

5.2 Retrieval Results

Table 5.4 shows the results on the main retrieval metrics used to evaluate the performance of the Claim Assistant. These metrics help us understand how often the system retrieves the correct articles, how well it ranks them, and whether retrieved articles are justified.

Table 5.4: Retrieval Results on Merged Dataset (percentages). *Arrows indicate whether higher or lower values are preferable: ↑ means higher is better; ↓ means lower is better.*

Metric	**Aegon**	**a.s.r.**
Retrieval Accuracy (Exact) ↑	43.88	55.10
Extended Accuracy (Exact + Prefix) ↑	46.94	72.45
Accuracy@1 ↑	9.18	46.94
Recall@1 ↑	8.91	44.23
Precision@1 ↑	9.18	46.94
Accuracy@3 ↑	14.29	67.35
Recall@3 ↑	13.86	63.46
Precision@3 ↑	4.76	22.45
F1@3 ↑	7.14	33.06
Mean Reciprocal Rank (MRR) ↑	11.73	56.46
Mean Average Precision (MAP) ↑	11.73	53.40
Coverage (No Match) ↓	53.06	27.55
Misleading Rate (Exact Not Justified) ↓	4.08	0.00
Extended Misleading Rate ↓	5.10	0.00
False Positive Rate (Exact) ↓	9.30	0.00
False Positive Rate (Extended) ↓	10.87	0.00
Justification Rate (All Matches) ↑	53.06	87.76
Misleading Coverage ↓	41.84	12.24

To start with, the Claim Assistant system retrieves the correct articles more frequently and ranks them higher for the a.s.r. dataset. For example, Retrieval Accuracy (Exact) is 55.10% for the a.s.r., compared to 43.88% for the Aegon dataset. When prefix matches are included, Extended Accuracy increases to 72.45% for a.s.r., but remains lower for Aegon at 46.94%. These metrics indicate whether the final article selected by the GPT model is an

exact or prefix match to the ground truth. The following metrics, including Accuracy@1 and Accuracy@3, instead measure whether the correct article appears in the retrieved list before the system's generation step. Additionally, unlike Extended Accuracy, these ranking metrics only count exact matches and do not consider prefix matches.

Accuracy@1, for instance, provides additional insight by measuring how often the correct article is ranked in the first position. In contrast, Retrieval Accuracy reflects whether the article selected by the system as the most relevant is correct, regardless of its rank in the retrieved list. This explains why Accuracy@1 is lower than Retrieval Accuracy (Exact), as it is 46.94% for the a.s.r. dataset and only 9.18% for the Aegon dataset.

When moving from only the first position to the top three, retrieval performance improves for both datasets. When evaluated on the a.s.r. dataset, the Claim Assistant reaches an Accuracy@3 of 67.35%, meaning that in most cases, the correct article is among the first three results. However, the Aegon dataset reaches only 14.29%, which shows that even with a more flexible threshold, relevant articles are often missing.

Recall and Precision scores offer further detail on ranking quality. The system achieves a Recall@1 of 44.23% for a.s.r., indicating that in nearly half of the queries, at least one of the correct articles appears in the first position. Recall@3 increases to 63.46%, showing that when the top three results are considered, the system retrieves relevant content more often.

The system achieves a Precision@1 of 46.94%, meaning that almost half of all top-ranked articles are correct. However, Precision@3 decreases to 22.45%, because even though the system retrieves more correct articles when expanding from one to three positions, the total number of retrieved items also increases, leading to a lower precision value.

In comparison, the Claim Assistant achieves a Recall@1 of 8.91% for the Aegon dataset, indicating that correct articles are found first in only a small fraction of queries. Recall@3 increases slightly to 13.86%. Precision@1 for Aegon is 9.18%, and Precision@3 remains low at 4.76%.

F1@3 provides a balanced view by combining precision and recall into a single score for the top three retrieved items. The system reaches an F1@3 score of 33.06% for a.s.r., reflecting a relatively good balance between including relevant articles and limiting incorrect ones. In contrast, the score for Aegon is much lower at 7.14%, confirming that even when more positions are considered, the retrieval remains inconsistent.

Ranking quality is also reflected in the Mean Reciprocal Rank (MRR) and Mean Average Precision (MAP) scores. The system reaches an MRR of 56.46% and a MAP of 53.40% for a.s.r., indicating that relevant articles are usually retrieved early and ranked consistently high. In contrast, the system achieves much lower scores for Aegon, with both MRR and MAP at 11.73%, confirming that correct articles often appear later in the list or are missing entirely.

Coverage (No Match) complements retrieval accuracy by showing how often none of the retrieved articles matches the ground truth at all. The system achieves a relatively low rate of 27.55% for a.s.r., while for Aegon it reaches 53.06%, indicating that the system often fails to retrieve any relevant content.

Misleading Rate captures how often the system retrieves articles that appear correct but are not justified by the knowledge base. The system reports both the exact and the extended misleading rates as 0% for a.s.r., indicating that there are no such incorrect matches. In contrast, for Aegon, the system has a small number of misleading results, with an Exact Misleading Rate of 4.08% and an Extended Misleading Rate of 5.10%, showing occasional retrieval of articles that do not correspond to the expected content.

False Positive Rate provides additional detail by indicating how often retrieved articles are labeled as correct matches but are actually unjustified. The system reports a 0% false positive rate for a.s.r. in both Exact and Extended cases, confirming that no false positives occur. For Aegon, however, these rates are higher, with an Exact False Positive Rate of 9.30% and an Extended False Positive Rate of 10.87%.

Justification Rate shows how often the retrieved articles are supported by

evidence from the knowledge base. The system achieves a high justification rate of 87.76% for a.s.r., meaning that most retrieved matches can be justified. In contrast, the system achieves only 53.06% for Aegon, suggesting that nearly half of its retrieved content lacks proper justification.

Finally, Misleading Coverage highlights the share of queries where no correct article is retrieved and none of the results are justified. This is observed in 41.84% of Aegon's cases, compared to only 12.24% for a.s.r.

5.3 Generation Results

For generation, we report seven metrics that capture lexical and semantic overlap between the system output and gold-standard answers: Cosine Similarity, Jaccard, Edit Distance, BLEU, and ROUGE (1, 2, L). Table 5.5 shows the mean scores over all queries.

Table 5.5: Generation Results on Merged Dataset. *Arrows indicate whether higher or lower values are preferable: ↑ means higher is better; ↓ means lower is better.*

Metric	**Aegon**	**a.s.r.**
Cosine Similarity ↑	0.3417	0.3663
Jaccard Similarity ↑	0.0478	0.0553
Edit Distance ↓	125.95	116.39
BLEU Score ↑	0.0063	0.0120
ROUGE-1 ↑	0.0985	0.1054
ROUGE-2 ↑	0.0336	0.0397
ROUGE-L ↑	0.0896	0.0980

In general, absolute scores are rather low, indicating that the generated answers often had limited overlap with the reference texts.

Cosine Similarity and Jaccard Similarity capture the degree of shared terms and semantic proximity. In these metrics, the system achieves slightly higher values for a.s.r. (0.3663 cosine and 0.0553 Jaccard) than for Aegon (0.3417 and 0.0478), suggesting that its outputs are somewhat closer in vocabulary and meaning to the ground truth. This could reflect either more consistent wording or more accurate selection of relevant passages.

In the BLEU and ROUGE metrics, which measure n-gram overlap and structural similarity, the system performs better for a.s.r. The BLEU score for a.s.r. is 0.0120, nearly double that of Aegon (0.0063). Although these values are low in absolute terms, they still indicate relatively more frequent matches of short phrases. ROUGE-1 and ROUGE-L follow the same trend, with the system reaching 0.1054 and 0.0980 for a.s.r., compared to 0.0985 and 0.0896 for Aegon. ROUGE-2, which focuses on bigram overlap, also favors a.s.r. (0.0397) over Aegon (0.0336), showing that slightly more consecutive word pairs are correctly reproduced.

Edit Distance provides a complementary perspective by quantifying how many text operations are needed to transform the system output into the gold answer. Since lower scores are better for this metric, the system achieves a smaller distance for a.s.r. (116.39) than for Aegon (125.95), indicating marginally closer outputs in terms of sequence alignment.

Taken together, these results suggest that the system generates answers for a.s.r. that are more similar to the references. This may be linked to the higher retrieval performance discussed earlier, since retrieving more relevant and justified articles likely provides better input for generation. However, the generally low scores across all metrics also highlight that the model does not consistently produce accurate textual outputs for either provider.

5.3.1 Test Dataset Results

In addition to the main evaluation, we also test the system on a smaller dataset with 9 real-world queries that were labeled by domain experts. Table 5.6 shows the retrieval metrics calculated for this test set.

Table 5.6: Retrieval Results on the Test Dataset (percentages). *Arrows indicate whether higher or lower values are preferable: ↑ means higher is better; ↓ means lower is better.*

Metric	**Aegon**	**a.s.r.**
Retrieval Accuracy (Exact) ↑	55.56	100.00
Extended Accuracy (Exact + Prefix) ↑	55.56	100.00
Accuracy@1 ↑	11.11	55.56
Recall@1 ↑	11.11	55.56
Precision@1 ↑	11.11	55.56
Accuracy@3 ↑	11.11	66.67
Recall@3 ↑	11.11	66.67
Precision@3 ↑	3.70	22.22
F1@3 ↑	5.56	33.33
Mean Reciprocal Rank (MRR) ↑	11.11	61.11
Mean Average Precision (MAP) ↑	11.11	61.11
Coverage (No Match) ↓	44.44	0.00
Misleading Rate (Exact Not Justified) ↓	0.00	0.00
Extended Misleading Rate ↓	0.00	0.00
False Positive Rate (Exact) ↓	0.00	0.00
False Positive Rate (Extended) ↓	0.00	0.00
Justification Rate (All Matches) ↑	55.56	100.00
Misleading Coverage ↓	44.44	0.00

Overall, these results show that the system performs very well for the a.s.r. dataset. It achieves 100% retrieval accuracy, and all retrieved articles are justified. For the Aegon dataset, retrieval performance is consistently lower across most metrics.

For instance, the system achieves a Recall@1 of 55.56% for a.s.r., indicating that in more than half of the queries, the correct article appears in the first position. In contrast, Recall@1 is only 11.11% for Aegon, showing that relevant articles rarely appear first.

When considering the top three results, Recall@3 increases to 66.67% for a.s.r., but stays at 11.11% for Aegon. Similarly, Precision@3 is higher for a.s.r. (22.22%) compared to Aegon (3.70%). As expected, the system achieves higher Mean Reciprocal Rank (MRR) and Mean Average Precision (MAP) for

a.s.r. (both 61.11%) than for Aegon (both 11.11%).

Coverage is a critical indicator. For a.s.r., all queries contain at least one correct match (Coverage = 0%). For Aegon, 44% of queries contain no correct match at all. These "no match" cases also lead to a Misleading Coverage of 44% for Aegon, meaning that in four queries, the system retrieves nothing correct and provides no justification.

No misleading retrievals are observed in this test set. Both datasets have 0% in Misleading Rate and Extended Misleading Rate, which is positive. Since no misleading retrievals means that all predictions marked as correct are actually justified by the knowledge base. There are no cases where the system confidently retrieves an unrelated article. Justification Rate is 100% for the a.s.r. dataset, showing that all retrieved articles are supported by the knowledge base. For the Aegon dataset, only 55% of the matches are justified.

In summary, the test dataset confirms the same pattern seen in the main dataset: retrieval performance for the a.s.r. dataset is more accurate and complete, while results for the Aegon dataset show lower accuracy, lower ranking scores, and significant coverage gaps.

5.3.2 Test Dataset Generation Results

For the generation metrics on the same test dataset of 9 queries. Table 5.7 shows the average scores for each metric.

Table 5.7: Generation Results on the Test Dataset (averages). *Arrows indicate whether higher or lower values are preferable: ↑ means higher is better; ↓ means lower is better.*

Metric	**Aegon**	**a.s.r.**
Cosine Similarity ↑	0.3092	0.2960
Jaccard Similarity ↑	0.0769	0.0534
Edit Distance ↓	96.56	140.78
BLEU Score ↑	0.0118	0.0042
ROUGE-1 ↑	0.1133	0.0917
ROUGE-2 ↑	0.0246	0.0106
ROUGE-L ↑	0.1019	0.0917

Overall, these generation results show that the system achieves slightly higher similarity scores in most metrics for the Aegon dataset. Interestingly, the strong retrieval performance of the system on the a.s.r. dataset does not directly translate into better generation scores. This suggests that other factors, such as the length and phrasing of gold answers, also influence generation outcomes. Thus, strong retrieval does not always guarantee high similarity scores when the references are short.

For example, the RAG system achieves higher BLEU (0.0118) and ROUGE-1 (0.1133) for the Aegon dataset, compared to 0.0042 and 0.0917 for the a.s.r. dataset. Cosine similarity is also slightly higher for the Aegon dataset (0.3092) than for the a.s.r. dataset (0.2960).

One reason for this difference is that the system's answers for the a.s.r. dataset are often longer and include more policy details, while the gold answers are short. This increases the edit distance for the a.s.r. dataset (140.78), showing that more text changes are needed to reach the reference. In contrast, the answers for the Aegon dataset sometimes use shorter sentences that match more words from the reference answer.

However, all scores remain low in absolute terms. This is mainly because the gold answers are often simple yes/no statements, and the system outputs have different wording and more explanations. This situation lowered BLEU and ROUGE scores, even though the overall meaning of the answers often remain correct.

Overall, these results show that for both insurers, the Claim Assistant produces answers that are semantically similar but not lexically identical to the gold answers.

5.4 Summary of the Results

To sum up, the evaluation shows that retrieval performance is consistently stronger for the a.s.r. dataset across both the main dataset and the test dataset. In the main dataset, the system achieves higher exact retrieval accuracy (55%) and better ranking metrics for the a.s.r. dataset compared to the Aegon

dataset. In the test dataset, the system retrieves the correct articles in all queries for the a.s.r. dataset, while 44% of Aegon queries have no matches. This gap suggests that retrieval for the a.s.r. dataset is more reliable and complete.

Generation metrics are more mixed. In the main dataset, the system's answers for the a.s.r. dataset show slightly higher similarity to the gold answers. In the test dataset, the system achieves higher average similarity scores for the Aegon dataset, especially in BLEU and ROUGE, because the generated texts often contain more words from the reference answers. In contrast, the answers for the a.s.r. dataset are sometimes shorter and more standard (for example, just "Ja" or "Nee"), which reduces the lexical overlap. However, all generation scores remain low overall. This is mainly because the reference answers are often short yes/no statements, while the model outputs include longer explanations.

These findings confirm that retrieval accuracy has a strong influence on generation quality. When the retrieved articles are correct and well-ranked, the generated answers are more likely to match the expected content. The results also show that content for both insurers can be retrieved and summarized effectively, but the system more often correctly identifies and ranks policy clauses from the a.s.r. dataset. This difference may be related to better formatting, chunking, or embedding alignment in the a.s.r. knowledge base.

Finally, the example query demonstrates that strict article matching can significantly impact ranking metrics. Even when the content is relevant and justified by the knowledge base, the metrics require exact article numbers to award a full score. This shows that combining strict metrics with justification checks provides a more complete view of retrieval quality.

6. Conclusion

This study investigated whether deterministic, non-LLM evaluation methods can reliably evaluate the retrieval and generation quality of a RAG system in a regulated insurance domain. The findings support the viability of this approach.

By applying a structured framework of 17 retrieval metrics and 7 generation metrics, we are able to measure system performance in detail without using black-box large language models for scoring. The results show that the Claim Assistant retrieves correct clauses more frequently and ranks them higher when working with policy documents from the a.s.r. dataset compared to those from the Aegon dataset. While the system achieves 55.1% exact retrieval accuracy and 72.45% extended accuracy for the a.s.r. in the main evaluation, retrieval performance is lower for the Aegon, with significant coverage gaps and lower ranking metrics.

For generation, similarity metrics show mixed results. In the main dataset, the system's answers for the a.s.r. dataset are slightly more similar to the gold answers. In the test dataset, the Claim Assistant's answers for the Aegon dataset achieve higher BLEU and ROUGE scores, mainly because the generated texts sometimes use wording closer to the reference answers. However, both datasets contain many semantically correct but lexically different outputs, which lowers BLEU and ROUGE values overall.

Together, these results confirm that a transparent, reproducible evaluation can be implemented using deterministic metrics. This approach helps identify which parts of the retrieval pipeline need improvement, such as clause ranking and coverage, without relying on subjective human scoring or additional LLMs.

6.1 Discussion

The main implication of this work is that insurance companies can use similar metric-based evaluation pipelines to monitor and improve RAG systems over time. Because all metrics in this study are deterministic and reproducible, it becomes possible to compare performance across different model versions, policy updates, and new user questions. This helps insurers understand where the system performs well and where improvements are needed.

One important insight is that strict evaluation rules, like requiring exact article matches, can sometimes lower the scores even if the retrieved content is still relevant. For example, in many Aegon cases, the system retrieved an article that included the correct sub-article but did not exactly match the article number. While this reduces ranking metrics, it does not always mean the answer is wrong. This shows that combining strict metrics with justification checks gives a clearer picture of retrieval quality.

From an ethical perspective, using transparent, non-LLM metrics reduces the risk of hidden bias or subjective judgments that can happen with black-box model scoring[10]. In regulated fields like insurance, this transparency is important to make sure results are fair, explainable, and compliant with legal standards. Reproducible metrics also help auditors and other stakeholders understand why certain answers were selected.

Overall, this project shows that statistical evaluation of RAG systems can be done in a clear and robust way. The approach used here gives insurers a base to improve retrieval and generation quality in a fair and measurable manner.

6.1.1 Limitations

This study has some limitations. First, it only includes two products (Inboedel and Reis insurance). Woonhuis insurance was not evaluated because there was no labeled ground truth data. So the results may not apply to all products.

Second, the evaluation only looked at the top-3 retrieved articles and the final generated answer. Other model outputs, like retrieval scores or intermediate steps, were not available. These could have given more details about how the system makes decisions.

Third, the test set had only 9 real user queries. Although this adds realism, the small number means the findings should be interpreted carefully. Bigger and more diverse datasets would help support stronger conclusions.

6.1.2 Ethical Implication and Considerations

Using AI tools in insurance comes with ethical risks. While the Claim Assistant helps users find and compare coverage, its answers are not legal advice. There is always a chance that users misunderstand the generated information, especially in complex cases.

Fairness also matters. The system uses metadata like customer type and coverage level to give more relevant results. However, care must be taken to avoid bias or giving some users better answers than others.

Finally, the study respects privacy. All queries were anonymous and contained no personal data, in line with the General Data Protection Regulation (GDPR) [1] and the European Statistics Code of Practice (Eurostat Code) [2], which are described in more detail below.

Alignment with the GDPR The GDPR establishes clear requirements for how personal data must be collected, processed, and stored within the European Union. In this study, all queries were fully anonymised and contained no identifiers. The evaluation relied only on aggregated metadata and generated outputs, ensuring compliance with the GDPR principles of data minimisation, purpose limitation, and transparency. No personal or sensitive information was processed at any stage.

[1] `https://eur-lex.europa.eu/eli/reg/2016/679/oj`

[2] `https://ec.europa.eu/eurostat/web/quality/european-quality-standards/european-statistics-code-of-practice`

Alignment with the European Statistics Code of Practice This research follows the *Eurostat Code* which sets out principles for quality, transparency, and reliability in the production of official statistics across the European Union[1]. The evaluation process in this study was designed to be reproducible and transparent, in line with the Eurostat Code, which can be found in Appendix E for a detailed review.

Instead of using human annotators or language models for evaluation, we created deterministic scripts that always produce the same results. This improves consistency because any user can repeat the evaluation and obtain identical outputs. All metrics are defined in detail, so it is easy to see how scores were calculated. Specifically, this work supports:

- **Principle 4: Commitment to Quality**, which requires statistical authorities to systematically identify strengths and weaknesses and to continuously improve processes and outputs. In line with this principle, all evaluation steps were documented and reproducibility was prioritised.
- **Principle 8: Appropriate Statistical Procedures**, which states that suitable and transparent procedures should be applied throughout all statistical processes to ensure quality. To meet this standard, established metrics such as Recall@k, MRR, BLEU, and Edit Distance were used.
- **Principle 12: Accuracy and Reliability**, which emphasises that statistics should accurately portray reality and be produced in a trustworthy manner. This includes the expectation that data, intermediate results, and outputs are regularly assessed and validated. The evaluation framework applied in this study reflects this principle, as it relies on deterministic, non-LLM methods to transparently and objectively assess retrieval and generation quality.

These choices help make the evaluation transparent, traceable, and easier to audit by other researchers or stakeholders.

6.1.3 Future Work

There are several promising directions for future research and system improvement. First, the current evaluation framework focuses on lexical and semantic similarity metrics but does not explicitly differentiate answer types. Given that many insurance questions, especially those involving coverage, are effectively binary (i.e., Yes/No), incorporating an additional evaluation module that measures agreement on binary response classification could provide more targeted feedback. Such a module would evaluate whether both system and ground truth responses imply the same binary decision, even if their wording differs.

Second, as highlighted by Diaz, Mitra, and Craswell (2016) [14], performance in retrieval-based systems can be improved by training query-specific or domain-specific embeddings. Although this study relies on OpenAI's `text-embedding-ada-003`, future work could experiment with training custom Word2Vec [3] or Sentence-BERT [4] models directly on Dutch insurance corpora. However, in this study, we chose to evaluate the system without retraining embeddings, leaving those more specialized embedding methods for future work. These experiments could be valuable in long-term scenarios where retraining costs are acceptable.

Finally, expanding the labeled datasets to include more products and question types would help future studies. A standard annotation approach could make it easier to track progress over time.

Altogether, these findings demonstrate that deterministic methods can effectively evaluate RAG systems in sensitive domains such as insurance. The approach developed in this study offers a reproducible framework that other researchers can adapt to monitor system performance over time. As retrieval and generation models continue to evolve, combining robust quantitative evaluation with domain-specific expertise will be essential to ensure that RAG systems remain accurate and aligned with regulatory requirements.

[3] `https://radimrehurek.com/gensim/models/word2vec.html`
[4] `https://www.sbert.net/`

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Code and Reproducibility Statement

All code and scripts used to generate the results in this thesis, including all relevant package versions and instructions, are available in the following GitHub repository:

`https://github.com/suezengin/nonllm-rag-evaluation`

This repository contains the complete evaluation pipeline described in Chapter 4, so that all analyses can be reproduced step by step.

Generative AI Statement

GPT-4-o (OpenAI) was used to assist with editing, formatting, and reordering text, creating example tables, and designing code templates for metric calculations. It was not used to generate domain knowledge, label data, or produce experimental results. All core analysis, validation, and interpretation were performed manually. Therefore, no generative AI system contributes intellectual property to this work. For a detailed overview of these uses, see Appendix D.

Appendices

A. System Pipeline

A.1 System Pipeline

For Claim Assistant, inference process is initiated when online Azure Machine Learning (Azure ML) endpoint gets a request from the web app, and a trained RAG model is used to generate outputs and sends the response back to the client. Thus, evaluating this process requires comprehensive approaches across different parts of the system.

Each system component, document chunking, embedding, indexing, and inference, is implemented as a separate Azure ML component, enabling version control, reproducibility, and scalability. The subsequent subsections break down each component of the pipeline, from document chunking to vector indexing, retrieval, and generation, highlighting their individual roles and integration within the overall RAG system.

Document Chunking and Preprocessing

Long insurance policy texts are first broken into semantically coherent chunks to enable accurate semantic search. The script `chunk_texts.py` processes raw `.jsonl` files by invoking a `Chunkinator` class, which reads each document applies recursive splitting via LangChain's `RecursiveCharacterTextSplitter` (or, optionally, fixed-character splitting), and writes segments back to disk in chunked `.jsonl` format. Each chunk retains its original metadata (e.g., product type, article number) so that downstream retrieval remains context-aware. This pipeline outputs segmented units optimized for embedding and a vector index based on **FAISS** (Facebook AI Similarity Search) an open-source library designed for efficient similarity search over dense vector representations.

Vector Index Construction and Registration

A modular Azure ML pipeline automates transforming chunks into a searchable format. First, the `chunk_texts` component splits raw `.jsonl` files into segments using a sliding window defined by `chunk_size` and `chunk_overlap`. These chunks feed into `create_embeddings` function, which calls OpenAI's `text-embedding-3-large` to produce dense vectors. The `create_faiss_index` component builds a FAISS index over these embeddings. Then, `combine_jsonl_input` merges the index with original chunk files, and `register_vector_index` registers the combined asset as a versioned `MLIndex` in Azure ML. All steps are defined in a single `pipeline.yml` for reproducibility and version control.

Vector Database Construction and Access

Once built and registered, the FAISS index is stored in Azure Blob Storage as a versioned data asset. The `VectorDB` class (in `vector_db.py`) leverages Azure ML's `MLClient` to fetch this index and wraps it with LangChain's vectorstore abstraction. `VectorDB` exposes methods such as `query_index()` for similarity search (top-k=3 by default) and `filter_responses_by_multiple_metadata()` to filter chunks by metadata (product, coverage, optional insurances). It also loads original `.jsonl` files into memory so that full article texts can be reconstructed during prompt generation. For local development, `vector_db_downloader.py` securely downloads the FAISS index and chunk data from Blob Storage for consistent behavior. Overall, these components work together ensuring access to document chunks.

Semantic Retrieval Logic

The `VectorSearch` class (in `vector_search.py`) orchestrates query processing. First, `classify_question()` sends the user's query and selected product metadata to GPT using a system prompt that lists insurance categories (e.g., baggage, cancellation, medical costs). Based on classification, and whether optional coverages apply, the system decides which clauses to include or exclude. This classification affects the metadata filters applied during retrieval.

For example, if a question is classified under "cancellation" but the user has not opted for cancellation coverage, the corresponding clauses are excluded.

Next, `search_through_vdb()` queries the FAISS index (via `query_index()`), retrieves top-ranked chunks by cosine similarity, and filters them by metadata defined by key-value pairs reflecting the product name, product type, coverage level (e.g., Basis, Allrisk), and optional supplementary insurances. Then, `choose_article_numbers()` selects which article numbers to include, prioritizing specific chapters per insurer (e.g., chapters 2 and 3 for Aegon, chapter 3 for a.s.r.), subject to a configurable limit to avoid token overflow.

Finally, `get_prompt()` reconstructs full article texts from the JSONL dictionary and formats them into a system prompt with explicit instructions for GPT. The prompt and user query are passed to GPT via `get_chat_completions()`, which generates the answer. Thus, `VectorSearch` serves as the main controller that integrates vector-based document retrieval with GPT-based answer generation in one pipeline.

Language Model Integration

All interaction with Azure OpenAI is handled by the `GPTAzureOpenAI` class in `gpt.py`. It handles both embedding generation (for retrieval) and promt-based chat completion (for generation), serving as the core GPT interface in a well-designed RAG pipeline.

Initialized with deployment details and credentials, it provides `generate_embedding()`, used during index construction to call `text-embedding-ada-002` and `chat_completions()`, used during inference to send combined system and user prompts to GPT. The prompts instruct GPT to return answers in a predefined JSON format: `Antwoord-Answer`, `Citaat-Cited clause`, `Artikel nummer-Article number`. If GPT's response is malformed, a retry mechanism issues a corrective prompt to reformat output into valid JSON.

In scenarios involving comparisons between different insurance providers, such as Aegon and a.s.r., the method `comparison_aegon_asr()` is used. This method constructs a prompt containing both answers and asks the language model to identify and summarize similarities and differences. The final

output is returned in markdown bullet-point format, enhancing clarity and readability for the end user. Thus, this function eliminates the need for users to manually read and compare policy clauses by leveraging GPT to analyze and summarize the differences between responses. By comparing the answers provided by two different insurance providers to the same question, the system offers a clear overview that enables customers to make more informed decisions. This also reduces the user's cognitive load by automatically presenting clause-level differences between policies.

Although implemented in a separate file, the GPT class is tightly coupled to the retrieval pipeline, and its methods are invoked at multiple points during query classification, clause selection, and response generation. As such, language model integration is not isolated but embedded within a tightly interdependent workflow.

Together, these methods allow the system to produce fluent, structured, and grounded responses that are tailored to the specific policy content retrieved during the search phase.

Dataset Integration for Inference

To enable seamless downstream use of the vector database and its associated chunked policy texts, the system includes a dedicated integration step. The script `combine_jsonl_input.py` is responsible for merging two essential components required during inference and evaluation: the FAISS index directory and the `.jsonl` directory containing preprocessed text chunks. Defined as an Azure ML component via YAML, it accepts `index_uri`, `jsonl_data_asset`, and `output_uri` (all as `uri_folder`). During execution, it copies index files and places chunked texts under a `jsonl/` subdirectory in the output folder. Running inside a containerized environment (`llm-rag-embeddings`) ensures consistency and reproducibility across development and production.

Request Handling and Inference Trigger

The entry point for handling user's requests is the `score.py` script. When the endpoint is initialized, the `init()` function loads the necessary components,

such as the vector database and the GPT model interface, using environment variables. Each individual inference request is handled by the `run()` function. It reads the input, delegates the request to the `SearchStrategies.run()` method, and returns the structured response to the user.

Query Processing and Comparison Strategy

The `search_strategies.py` module ties everything together. Its `run()` method receives a user question plus metadata (insurance product, coverage, optional clauses) and chooses a product specific strategy:

- Determines the product type (e.g., Aegon, a.s.r.) and relevant clause filters.
- Performs semantic retrieval for each insurer.
- Constructs prompts using top-k articles.
- Generates answers via GPT for both insurers.
- Invokes a GPT-based comparison routine to identify similarities and differences.
- The output is returned in a structured JSON format suitable for UI rendering and evaluation.

The final output is a structured dictionary containing each insurer's answer, the comparative analysis, and diagnostic metadata (e.g., classification confidence). This modular design allows flexible strategy logic while keeping retrieval and generation tightly integrated into a unified RAG workflow.

B. Data Example Tables

B.1 Example of Merged Evaluation Dataset in Dutch

Table B.1: Voorbeeldrij uit de samengevoegde evaluatiedataset (oorspronkelijke Nederlandse versie)

Kolom	**Waarde**
question	Ik heb een hersenschudding, ik wil annuleren, Is dit verzekerd?
product	Doorlopende reisverzekering
dekking	Basis
type_klant	Adviseur
polis_versie	
comparison_response_antwoord	Beide verzekeraars erkennen een hersenschudding als mogelijke reden voor annulering bij ernstige ziekte.
aegon_system_answer	Ja, een hersenschudding valt onder een ernstige ziekte, wat een geldige reden is voor annulering volgens de verzekeringsvoorwaarden.
asr_system_answer	Ja, als u ernstig ziek wordt of een ernstig ongeval krijgt, waaronder een hersenschudding kan vallen..
top_3_aegon	['2.5', '2.6', '2.2']
top_3_asr	['3.10', '4.4', '3.1']
system_article_ids_aegon	['2.5.1']
system_article_ids_asr	['3.10']
gt_article_ids_aegon	['2.5.1']
gt_article_ids_asr	['3.10']
aegon_gold_answer	ja, in punt 1 staat "ernstige ziekte" in hoofdstuk 6 staat bij de begrippenlijst wat wij bedoelen met ernstige ziekte, hier wordt hersenschudding niet genoemd maar we hebben besloten om de lijst niet meer te volgen
asr_gold_answer	ja
gt_source_aegon	aegon_reis
gt_source_asr	asr_vp_dr_2024
kb_source_aegon	aegon_reis
kb_source_asr	asr_vp_dr_2024
sub_mentions_in_content	[]
mentions_other_articles	['3.1']

B.2 Enriched Row Examples

Table B.2: Voorbeeld verrijkte rij – Inboedel (originele Nederlandse versie)

Vraag	Artikel	Antwoord	GT Source	Product	Klanttype	Dekking
Er is een bal door mijn ruit gegaan. Wordt dit vergoed?	3.3	Nee, tenzij aanvullend verzekerd voor glas.	a.s.r._ikz_basis	Inboedel	Direct	Basis

Table B.3: Voorbeeld verrijkte rij – Reis (originele Nederlandse versie)

Vraag	Antwoord	Artikel	GT Source	Product	Klanttype	Dekking
Ik heb een hersenschudding, ik wil annuleren, Is dit verzekerd?	ja, in punt 1 staat "ernstige ziekte." In hoofdstuk 6 staat bij de begrippenlijst wat wij bedoelen met ernstige ziekte, hier wordt hersenschudding niet genoemd maar we hebben besloten om de lijst niet meer te volgen.	2.5.1	aegon_basis	Doorlopende Reisverzekering	(leeg)	Basis

B.3 Knowledge Base Article Examples

Table B.4: Voorbeeld extractie uit kennisbank (oorspronkelijke Nederlandse versie)

Veld	Inhoud
article_number	1.14

content	1.14 Wanneer begint en eindigt de verzekering? 1.14.1 Begint op de ingangsdatum van de polis. Schade voor deze datum is niet gedekt. 1.14.2 U kunt na het eerste jaar op elk moment opzeggen. Restitutie mogelijk. Extra dekkingen kunnen worden verwijderd of verlaagd. 1.14.3 Wij beëindigen direct bij fraude, verzwijging, wanbetaling of gebrek aan medewerking. 1.14.4 Wij beëindigen met opzegtermijn bij wanbetaling (zie 1.18), emigratie, verlenging of schadeclaims.
sub_mentions_-in_content	['1.14.1', '1.14.2', '1.14.3', '1.14.4']
mentions_other_-articles	['1.14.1', '1.14.2', '1.14.3', '1.14.4', '1.18']
kb_source	aegon_reis
kb_source_group	aegon

B.4 Pivot Format Examples for Product

Table B.5: Example Pivot Row for Inboedel – (English Translation)

a.s.r. IKZ	**basis**	
Question	**Article**	**Answer**
I dropped my phone. Is this covered?	Chapter 3.6	No, unless mobile electronics are additionally insured.

Table B.6: Voorbeeld van oorspronkelijke draaitabelstructuur voor Inboedel – `a.s.r. IKZ basis`

a.s.r. IKZ	**basis**	
Vraag	**Artikelnum**	**Verzekerd?**
Ik heb mijn telefoon laten vallen. Is dit verzekerd?	Hoofdstuk 3.6	Nee, tenzij mobiele elektronica is meeverzekerd.

Table B.7: Example Pivot Row for Reis – (English Translation)

Aegon	**basis**	
Question	**Article**	**Answer**
I have a concussion, I want to cancel, is this insured?	2.5.1 point 1	yes, in point 1 it says "serious illness"...

Table B.8: Voorbeeld draaitabel voor Reis – originele versie in het Nederlands

Aegon	**basis**	
Vraag	**Artikelnum**	**Verzekerd?**
Ik heb een hersenschudding, ik wil annuleren, Is dit verzekerd?	2.5.1	ja, in punt 1 staat "ernstige ziekte" in hoofdstuk 6.

C. Python Functions for Retrieval Evaluation

Evaluation Data Preparation Functions

extract_article_numbers()

```
import re
import ast
import pandas as pd

def extract_article_numbers(text):
    if text is None or (isinstance(text, float) and pd.isna(
       text)):
        return []
    if isinstance(text, list):
        return text
    try:
        parsed = ast.literal_eval(text)
        if isinstance(parsed, list):
            return [str(x).strip() for x in parsed if x]
    except:
        pass
    article_pattern = re.compile(r"\b\d+(?:\.\d+){1,2}\b")
    return article_pattern.findall(str(text))
```

preprocess_article_ids_fixed()

```
def preprocess_article_ids_fixed(df):
    df["system_article_ids_aegon"] = df["aegon_response_Artikel
       ␣nummer"].apply(extract_article_numbers)
    df["system_article_ids_asr"] = df["asr_response_Artikel␣
       nummer"].apply(extract_article_numbers)
```

```
    df["gt_article_ids_aegon"] = df["aegon_gold_article_id"].
        apply(extract_article_numbers)
    df["gt_article_ids_asr"] = df["asr_gold_article_id"].apply(
        extract_article_numbers)
    df["selected_articles_aegon"] = df["selected_articles_aegon
        "].apply(extract_article_numbers)
    df["top_3_aegon"] = df["top_3_aegon"].apply(
        extract_article_numbers)
    df["selected_articles_asr"] = df["selected_articles_asr"].
        apply(extract_article_numbers)
    df["top_3_asr"] = df["top_3_asr"].apply(
        extract_article_numbers)
    return df
```

has_gt_quality_issue()

```
def has_gt_quality_issue(article_ids):
    if not isinstance(article_ids, list):
        return True
    if not article_ids:
        return True
    for aid in article_ids:
        if not re.match(r"^\d+(\.\d+)*$", str(aid)):
            return True
    return False
```

classify_match()

```
def classify_match(system_articles, gold_articles):
    if not system_articles or not gold_articles:
        return "no_match"
    if any(sa == ga for sa in system_articles for ga in
        gold_articles):
        return "exact"
    if any(sa.startswith(ga) or ga.startswith(sa) for sa in
        system_articles for ga in gold_articles):
        return "prefix"
    return "no_match"
```

determine_kb_source_aegon()

```
def determine_kb_source_aegon(polis_versie, product):
    polis_versie = str(polis_versie).lower().strip()
    product = str(product).lower().strip()
    if "inboedel" in product:
        if "nieuw" in polis_versie:
            return "aegon_inboedel_nieuw"
        elif "oud" in polis_versie:
            return "aegon_inboedel_oud"
    if "doorlopende reisverzekering" in product:
        return "aegon_reis"
```

determine_kb_source_asr()

```
def determine_kb_source_asr(source_gt_asr):
    source_gt_asr = str(source_gt_asr).lower().strip()
    if "vp_dr_2024" in source_gt_asr:
        return "asr_vp_dr_2024"
    elif "ikz_allrisk" in source_gt_asr:
        return "asr_ikz_allrisk"
    elif "ikz_2018" in source_gt_asr:
        return "asr_ikz_2018"
    elif "advies_basis" in source_gt_asr:
        return "asr_advies_basis"
    return None
```

is_justified_by_kb()

```
def is_justified_by_kb(system_articles, gold_articles, kb_df,
   kb_source):
    if not kb_source:
        return False

    filtered_kb = kb_df[kb_df["kb_source"] == kb_source].copy()
    if filtered_kb.empty:
        return False

    def parse_list_column(col):
```

```
    return col.apply(lambda x: extract_article_numbers(x))

filtered_kb["article_number"] = filtered_kb["article_number
    "].astype(str)
filtered_kb["sub_mentions_in_content"] = parse_list_column(
    filtered_kb["sub_mentions_in_content"])
filtered_kb["mentions_other_articles"] = parse_list_column(
    filtered_kb["mentions_other_articles"])

kb_all_refs = []
for _, row in filtered_kb.iterrows():
    refs = set([row["article_number"]] + row["
        sub_mentions_in_content"] + row["
        mentions_other_articles"])
    kb_all_refs.append(refs)

flat_kb_articles = set().union(*kb_all_refs)
match_type = classify_match(system_articles, gold_articles)

if match_type == "exact":
    return any(x in flat_kb_articles for x in
        system_articles + gold_articles)

if match_type == "prefix":
    return any(
        any(sa in refs for sa in system_articles) and
        any(ga in refs for ga in gold_articles)
        for refs in kb_all_refs
    )

return any(
    any(sa in refs for sa in system_articles) and
    any(ga in refs for ga in gold_articles)
    for refs in kb_all_refs
)
```

Retrieval Evaluation Functions

compute_all_retrieval_metrics_dataframe()

```
def compute_all_retrieval_metrics_dataframe(df, source="asr", k
    =3):
     assert source in ["asr", "aegon"]
     match_col = f"match_type_{source}"
     justify_col = f"justified_by_kb_{source}"
     gt_col = f"gt_article_ids_{source}"
     retrieved_col = f"top_{k}_{source}"

     rows = []
     for _, row in df.iterrows():
         out = {}
         retrieved = row.get(retrieved_col, [])
         gt_ids = row.get(gt_col, [])
         if not isinstance(retrieved, list) or not isinstance(
            gt_ids, list) or len(gt_ids) == 0:
             out["precision_at_k"] = None
             out["recall_at_k"] = None
             out["f1_at_k"] = None
             out["accuracy_at_k"] = None
         else:
             gt_set = set(gt_ids)
             top_k = retrieved[:k]
             relevant = gt_set.intersection(top_k)

             prec = len(relevant) / k if k > 0 else 0
             rec = len(relevant) / len(gt_set) if len(gt_set) >
                0 else 0
             f1 = (2 * prec * rec / (prec + rec)) if (prec + rec
                ) > 0 else 0
             acc = 1 if len(relevant) > 0 else 0

             out["precision_at_k"] = prec
             out["recall_at_k"] = rec
             out["f1_at_k"] = f1
             out["accuracy_at_k"] = acc
```

```
            out["match_type"] = row.get(match_col, None)
            out["justified_by_kb"] = row.get(justify_col, None)
            rows.append(out)

    metrics_df = pd.DataFrame(rows)
    metrics_df.columns = [f"{source}_{col}" for col in
        metrics_df.columns]
    return metrics_df
```

compute_row_metrics_for_source()

```
def is_extended_match(match_type):
    return match_type in ["exact", "prefix"]

def compute_row_metrics_for_source(df, source="asr", k=3):
    match_col = f"match_type_{source}"
    justify_col = f"justified_by_kb_{source}"
    total_queries = len(df)

    exact_count = (df[match_col] == "exact").sum()
    extended_count = df[match_col].apply(is_extended_match).sum
        ()
    coverage_count = (df[match_col] == "no_match").sum()

    misleading_exact = df[(df[match_col] == "exact") & (~df[
        justify_col])].shape[0]
    misleading_extended = df[df[match_col].apply(
        is_extended_match) & (~df[justify_col])].shape[0]
    misleading_coverage_count = df[(df[match_col] == "no_match"
        ) & (~df[justify_col])].shape[0]

    metrics = {
        "retrieval_accuracy": (exact_count, exact_count /
            total_queries * 100),
        "extended_accuracy": (extended_count, extended_count /
            total_queries * 100),
        "coverage": (coverage_count, coverage_count /
            total_queries * 100),
```

```
        "justification_rate": (df[justify_col].sum(), df[
            justify_col].mean() * 100),
        "misleading_exact_rate": (misleading_exact,
            misleading_exact / exact_count * 100 if exact_count
            > 0 else 0),
        "misleading_extended_rate": (misleading_extended,
            misleading_extended / extended_count * 100 if
            extended_count > 0 else 0),
        "misleading_coverage_rate": (misleading_coverage_count,
             misleading_coverage_count / total_queries * 100),
    }
    return metrics
```

run_all_metrics_dual()

```
def run_all_metrics_dual(df, filter_by_gt_quality=True, k=3):
    if filter_by_gt_quality:
        df = df[
            (df["gt_quality_issue_asr"] != True) &
            (df["gt_quality_issue_aegon"] != True)
        ].copy()

    metrics_asr = compute_row_metrics_for_source(df, source="
        asr", k=k)
    metrics_aegon = compute_row_metrics_for_source(df, source="
        aegon", k=k)

    rows = []
    for metric_name in metrics_asr:
        asr_count, asr_pct = metrics_asr[metric_name]
        aegon_count, aegon_pct = metrics_aegon[metric_name]
        rows.append({
            "Metric": metric_name,
            "ASR␣Count": asr_count,
            "ASR␣%": f"{asr_pct:.2f}%",
            "AEGON␣Count": aegon_count,
            "AEGON␣%": f"{aegon_pct:.2f}%"
        })

    return pd.DataFrame(rows)
```

D. Use of Generative AI

Below is a concise overview of how generative AI tools were used in this thesis. The descriptions are provided to maintain transparency without disclosing full prompt texts. All examples shown here are illustrative.

Tool

ChatGPT (GPT-4-o, OpenAI)

Scope of Use

1. Table Preparation

Assistance with formatting Dutch-language tables for Overleaf.

Example Prompt (illustrative):

> Create LaTeX tables displaying Dutch queries, associated insurance metadata, and corresponding English translations. Ensure that the tables fit within A4 margins and use consistent font sizes.

2. Code Assistance

Help designing Python scripts for calculating retrieval and generation metrics.

Example Prompt (illustrative):

> Implement functions to compute Recall@k, MRR@k and MAP based on formulas from the literature review (pasting formulas), using specific columns, such as match_type, justified_by_kb, gt_article_ids, and top_k retrieved articles, with given mapping logic. Provide modular code that can be run in AzureML.

3. Text Drafting and Simplification

Reordering the sections to change the order of paragraphs for clarity and logical flow without changing the actual text.

Example Prompt (illustrative):

> Rewrite the result chapter to move the example query subsection before the retrieval evaluation result subsection.

4. Drafting Supporting Sections

Help in drafting non-technical supporting sections, such as acknowledgments and appendices.

Example Prompt (illustrative):

> Suggest a concise acknowledgment text thanking my scholarship program with an academic tone.

5. Footnote Creation

Help creating clickable footnotes in Overleaf.

Example Prompt (illustrative):

> How to create clickable footnote for FAISS in Overleaf?

6. Consistency Checks

Help checking consistency of terminology across the document.

Example Prompt (illustrative):

> Review my methodology section text to check if I consistently refer to Retrieval Accuracy as "Selection Accuracy" throughout.

7. Diagram Drafting Suggestions

Help with designing pipeline diagrams for the methodology chapter.

Example Prompt (illustrative):

> Can you suggest a TikZ diagram showing the Retrieval-Augmented Generation pipeline? The diagram should include labels like "Query Filtering/Reformulation," "Query Embedding," "Document Embedding and Indexing," "Similarity Search," and "Answer Generation." Use arrows to show the flow from the user query through retrieval and generation steps.

8. Data Transformation Assistance

Help designing code for unpivoting Excel datasets into a clause-level format.

Example Prompt (illustrative):

> How can I transform a wide-format Excel table containing multiple product versions as separate columns into a long-format table where each row corresponds to one question-product combination? I want to include metadata columns such as product, coverage type, and policy version, and keep the original question text and answer.

Note on Content Generation

Generative AI was not used to produce domain knowledge, label the dataset, or generate experimental results. Its role was limited to:

- Code structuring
- Rewriting text for clarity
- Formatting assistance

All final analysis, data preparation, and validation were performed manually.

E. European Statistics Code of Practice

EUROPEAN STATISTICS CODE OF PRACTICE

For the National Statistical Authorities and Eurostat (EU statistical authority)

Adopted by the
European Statistical System Committee

16th November 2017

eurostat

EUROPEAN STATISTICAL SYSTEM

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EUROPEAN STATISTICAL SYSTEM

Preamble

The European Statistics Code of Practice

The European Statistics Code of Practice is the cornerstone of the common quality framework of the European Statistical System. It is a self-regulatory instrument and is based on 16 Principles covering the institutional environment, statistical processes and statistical outputs. A set of indicators of best practices and standards for each of the Principles provides guidance and reference for reviewing the implementation of the Code of Practice, increasing transparency within the European Statistical System.

Statistical authorities, comprising the European Union Statistical Authority (Eurostat), the National Statistical Institutes and other national authorities responsible for the development, production and dissemination of European Statistics, commit themselves to adhere to the Code of Practice.

An independent Advisory Board, the European Statistical Governance Advisory Board[1] (ESGAB, the Board) provides an overview of the European Statistical System as regards the implementation of the Code of Practice. The Board analyses the implementation of the Code of Practice by the European Union Statistical Authority (Eurostat) and the European Statistical System as a whole every year; and advises on appropriate measures to facilitate the implementation of the Code of Practice, on its communication to users and data providers as well as on its possible updates.

This 2017 edition is the second revision of the Code of Practice, originally adopted in 2005. It aims at reflecting the latest changes and innovations in the development, production and dissemination of official statistics in the European Statistical System and beyond, such as emerging new data sources, use of new technologies, modernisation of the legal framework, and the results of the peer reviews on the implementation of the Code of Practice.

A Glossary explaining the main terms used in the Code of Practice is available online at http://ec.europa.eu/eurostat/web/quality/overview.

[1] European Statistical Governance Advisory Board: http://ec.europa.eu/eurostat/web/esgab/introduction

The common quality framework of the European Statistical System

The common quality framework of the European Statistical System is composed of the European Statistics Code of Practice, the Quality Assurance Framework of the European Statistical System and the general quality management principles (such as continuous interaction with users, commitment of leadership, partnership, staff satisfaction, continuous improvement, integration and harmonisation).

This self-regulatory common quality framework complements the extensive legal framework of the European Statistical System based on the Regulation (EC) No 223/2009 on European statistics which itself derives from the Treaty on the Functioning of the European Union. High-quality European statistics and services are therefore developed, produced and disseminated in a very robust legal and quality framework.

The Quality Declaration of the European Statistical System, included in the subsequent pages of this brochure, demonstrates the quality awareness that exists in the European Statistical System as well as the self-commitment of all its members to continuously developing, producing and disseminating high-quality European statistics and services in order to sustainably provide value to its users.

EUROPEAN
STATISTICAL
SYSTEM

Quality Declaration of the European Statistical System

The European Statistical System is a partnership in which Eurostat and the national statistical authorities of each EU Member State and EFTA country cooperate. Together, our mission is to provide independent high quality statistical information at European, national and regional levels and to make this information available to everyone for decision-making, research and debate.

The programme and priorities of European statistics are discussed and agreed by ESS members with the final decisions taken in a democratic manner according to European legislative procedures.

We operate under a strict legal regime[1] supplemented by a robust, world-class and self-regulatory quality framework, the backbone of which is the European Statistics Code of Practice[2]. Our compliance with the Code of Practice is periodically assessed by means of review mechanisms and strict follow-up of the improvement actions[3] identified.

[1] Treaty on the Functioning of the European Union: http://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:C:2016:202:FULL&from=EN and Regulation (EC) No 223/2009 on European statistics as amended by Regulation (EU) 2015/759: http://eur-lex.europa.eu/legal-content/en/TXT/PDF/?uri=CELEX:02009R0223-20150608&from=EN

[2] European Statistics Code of Practice: http://ec.europa.eu/eurostat/documents/3859598/5921861/KS-32-11-955-EN.PDF and Quality Assurance Framework of the ESS: http://ec.europa.eu/eurostat/documents/64157/4392716/ESS-QAF-V1-2final.pdf/bbf5970c-1adf-46c8-afc3-58ce177a0646

[3] First and second round of ESS peer reviews: http://ec.europa.eu/eurostat/web/quality/first-round-of-peer-reviews & http://ec.europa.eu/eurostat/web/quality/peer-reviews

We see quality as the basis of our competitive advantage in a world experiencing a growing trend of instant information which often lacks the necessary proof of quality. Our work is governed by professional independence, impartial treatment of all our users, objectivity, reliability, statistical confidentiality and cost-effectiveness. The development, production and dissemination of our statistics are based on sound methodologies, the best international standards and appropriate procedures that are well documented in a transparent manner. Our principles of quality are: relevance, accuracy, timeliness and punctuality, accessibility and clarity, as well as comparability and coherence.

We consistently strive to minimise the burden on our respondents, cultivate good cooperation with data providers and cooperate closely with stakeholder groups including the scientific communities.

We are committed to statistical excellence by systematically identifying our strengths and weaknesses, as well as related risks which we duly take into account by the continuous development of our common quality framework. By continuously modernising, innovating and compiling new indicators, we not only improve the quality of our products and services, but also attempt to anticipate emerging phenomena and needs with our users.

Heads of the National Statistical Institutes and Director-General of Eurostat

Institutional environment

Institutional and organisational factors have a significant influence on the effectiveness and credibility of a statistical authority developing, producing and disseminating European Statistics. The relevant Principles are professional independence, coordination and cooperation, mandate for data collection, adequacy of resources, quality commitment, statistical confidentiality, impartiality and objectivity.

PRINCIPLE 1

Professional Independence

Professional independence of statistical authorities from other policy, regulatory or administrative departments and bodies, as well as from private sector operators, ensures the credibility of European Statistics.

INDICATOR

1.1 The independence of the National Statistical Institutes and Eurostat from political and other external interference in developing, producing and disseminating statistics is specified in law and assured for other statistical authorities.

1.2 The heads of the National Statistical Institutes and of Eurostat and, where appropriate, the heads of other statistical authorities have sufficiently high hierarchical standing to ensure senior level access to policy authorities and administrative public bodies. They are of the highest professional calibre.

1.3 The heads of the National Statistical Institutes and of Eurostat and, where appropriate, the heads of other statistical authorities have responsibility for ensuring that statistics are developed, produced and disseminated in an independent manner.

1.4 The heads of the National Statistical Institutes and of Eurostat and, where appropriate, the heads of other statistical authorities have the sole responsibility for deciding on statistical methods, standards and procedures, and on the content and timing of statistical releases.

1.5 The statistical work programmes are published and periodic reports describe progress made.

1.6 Statistical releases are clearly distinguished and issued separately from political/policy statements.

1.7 The National Statistical Institutes and Eurostat and, where appropriate, other statistical authorities, comment publicly on statistical issues, including criticisms and misuses of statistics as far as considered suitable.

1.8 The procedures for the recruitment and appointment of the heads of the National Statistical Institutes and Eurostat and, where appropriate, the statistical heads of other statistical authorities, are transparent and based on professional criteria only. The reasons on the basis of which the incumbency can be terminated are specified in the legal framework. These cannot include reasons compromising professional or scientific independence.

PRINCIPLE 1bis

Coordination and cooperation

National Statistical Institutes and Eurostat ensure the coordination of all activities for the development, production and dissemination of European statistics at the level of the national statistical system and the European Statistical System, respectively. Statistical authorities actively cooperate within the partnership of the European Statistical System, so as to ensure the development, production and dissemination of European statistics.

INDICATOR

1bis.1 The National Statistical Institutes coordinate the statistical activities of all other national authorities that develop, produce and disseminate European statistics. They act in this regard as the sole contact point for Eurostat on statistical matters. Legislation and well defined and established procedures are in place for implementing the coordination role at both national and European levels.

1bis.2 National guidelines to ensure quality in the development, production and dissemination of European statistics within the national statistical system are produced by the Heads of the National Statistical Institutes, where necessary; their implementation is monitored and reviewed.

1bis.3 Statistical authorities continuously maintain and develop cooperation at various levels with each other and with the advisory bodies of the European Statistical System, as well as with the members of the European System of Central Banks, academic institutions and other international bodies, as appropriate.

PRINCIPLE 2

Mandate for Data Collection and Access to Data

Statistical authorities have a clear legal mandate to collect and access information from multiple data sources for European statistical purposes. Administrations, enterprises and households, and the public at large may be compelled by law to allow access to or deliver data for European statistical purposes at the request of statistical authorities.

INDICATOR

2.1 The mandate of the statistical authorities to collect and access information from multiple data sources for the development, production and dissemination of European Statistics is specified in law.

2.2 The statistical authorities are allowed by law to access administrative data, promptly and free of charge, and use them for statistical purposes. They are involved from the beginning in the design, development and discontinuation of administrative records, in order to make them more suitable for statistical purposes.

2.3 On the basis of a legal act, the statistical authorities may compel response to statistical surveys.

2.4 Access for statistical purposes to other data, such as privately held data, is facilitated, while ensuring statistical confidentiality and data protection.

PRINCIPLE 3

Adequacy of Resources

The resources available to statistical authorities are sufficient to meet European Statistics requirements.

INDICATOR

3.1 Human, financial and technical resources, adequate both in magnitude and in quality, are available to meet statistical needs.

3.2 The scope, detail and cost of statistics are commensurate with needs.

3.3 Procedures exist to assess and justify demands for new statistics against their cost.

3.4 Procedures exist to assess the continuing need for all statistics, to see if any can be discontinued or curtailed to free up resources.

PRINCIPLE 4

Commitment to Quality

Statistical authorities are committed to quality. They systematically and regularly identify strengths and weaknesses to continuously improve process and output quality.

INDICATOR

4.1 Quality policy is defined and made available to the public. An organisational structure and tools are in place to deal with quality management.

4.2 Procedures are in place to plan, monitor and improve the quality of the statistical processes, including the integration of data from multiple data sources.

4.3 Output quality is regularly monitored, assessed with regard to possible trade-offs, and reported according to the quality criteria for European Statistics.

4.4 There is a regular and thorough review of the key statistical outputs using also external experts where appropriate.

PRINCIPLE 5

Statistical Confidentiality and Data Protection

The privacy of data providers, the confidentiality of the information they provide, its use only for statistical purposes and the security of the data are absolutely guaranteed.

INDICATOR

5.1 Statistical confidentiality is guaranteed in law.

5.2 Staff sign legal confidentiality commitments on appointment.

5.3 Penalties are prescribed for any wilful breaches of statistical confidentiality.

5.4 Guidelines and instructions are provided to staff on the protection of statistical confidentiality throughout the statistical processes. The confidentiality policy is made known to the public.

5.5 The necessary regulatory, administrative, technical and organisational measures are in place to protect the security and integrity of statistical data and their transmission, in accordance with best practices, international standards, as well as European and national legislation.

5.6 Strict protocols apply to external users accessing statistical microdata for research purposes.

PRINCIPLE 6

Impartiality and Objectivity

Statistical authorities develop, produce and disseminate European Statistics respecting scientific independence and in an objective, professional and transparent manner in which all users are treated equitably.

INDICATOR

6.1 Statistics are compiled on an objective basis determined by statistical considerations.

6.2 Choices of data sources and statistical methods as well as decisions about the dissemination of statistics are based on statistical considerations.

6.3 Errors discovered in published statistics are corrected at the earliest possible date and publicised.

6.4 Information on data sources, methods and procedures used is publicly available.

6.5 Statistical release dates and times are pre-announced.

6.6 Advance notice is given on major revisions or changes in methodologies.

6.7 Statistical authorities independently decide on the time and content of statistical releases, while taking into account the goal of providing complete and timely statistical information. All users have equal access to statistical releases at the same time. Any privileged pre-release access to any outside user is limited, well-justified, controlled and publicised. In case of breach, pre-release arrangements are reviewed so as to ensure impartiality.

6.8 Statistical releases and statements made in press conferences are objective and non-partisan.

Statistical Processes

European and other international standards, guidelines and good practices are fully observed in the statistical processes used by the statistical authorities to develop, produce and disseminate European Statistics, while constantly striving for innovation. The credibility of the statistics is enhanced by a reputation for good management and efficiency. The relevant Principles are sound methodology, appropriate statistical procedures, non-excessive burden on respondents and cost effectiveness.

PRINCIPLE 7

Sound Methodology

Sound methodology underpins quality statistics. This requires adequate tools, procedures and expertise.

INDICATOR

7.1 The overall methodological framework used for European Statistics follows European and other international standards, guidelines, and good practices, while constantly striving for innovation.

7.2 Procedures are in place to ensure that standard concepts, definitions, classifications and other types of standards are consistently applied throughout the statistical authority.

7.3 The registers and frames used for European Statistics are regularly evaluated and adjusted if necessary in order to ensure high quality.

7.4 Detailed concordance exists between national classifications systems and the corresponding European systems.

7.5 Graduates in the relevant academic disciplines are recruited.

7.6 Statistical authorities implement a policy of continuous vocational training for their staff.

7.7 Statistical authorities maintain and develop cooperation with the scientific community to improve methodology, the effectiveness of the methods implemented and to promote better tools when feasible.

PRINCIPLE 8

Appropriate Statistical Procedures

INDICATOR Appropriate statistical procedures, implemented throughout the statistical processes, underpin quality statistics.

8.1 When European Statistics are based on administrative and other data, the definitions and concepts used for non-statistical purposes are a good approximation to those required for statistical purposes.

8.2 In the case of statistical surveys, questionnaires are systematically tested prior to the data collection.

8.3 Statistical processes are routinely monitored and revised as required.

8.4 Metadata related to statistical processes are managed throughout the statistical processes and disseminated, as appropriate.

8.5 Revisions follow standard, well-established and transparent procedures.

8.6 Agreements are made with holders of administrative and other data which set out their shared commitment to the use of these data for statistical purposes.

8.7 Statistical authorities co-operate with holders of administrative and other data in assuring data quality.

PRINCIPLE 9

Non-excessive Burden on Respondents

INDICATOR The response burden is proportionate to the needs of the users and is not excessive for respondents. The statistical authorities monitor the response burden and set targets for its reduction over time.

9.1 The range and detail of European Statistics demands is limited to what is absolutely necessary.

9.2 The response burden is spread as widely as possible over survey populations and monitored by the statistical authority.

9.3 The data sought from businesses is, as far as possible, readily available from their accounts and electronic means are used where possible to facilitate its return.

9.4 Administrative and other data sources are used whenever possible to avoid duplicating requests for data.

9.5 Data sharing and data integration, while adhering to confidentiality and data protection requirements, are promoted to minimise response burden.

9.6 Statistical authorities promote measures that enable the linking of data sources in order to minimise response burden.

PRINCIPLE 10

Cost Effectiveness

INDICATOR Resources are used effectively.

10.1 Internal and independent external measures monitor the statistical authority's use of resources.

10.2 The productivity potential of information and communication technology is being optimised for the statistical processes.

10.3 Proactive efforts are made to improve the statistical potential of administrative and other data sources and to limit recourse to direct surveys.

10.4 Statistical authorities promote, share and implement standardised solutions that increase effectiveness and efficiency.

Statistical Output

Available statistics meet users' needs. Statistics comply with the European quality standards and serve the needs of European institutions, governments, research institutions, business concerns and the public generally. Output quality is measured by the extent to which the statistics are relevant, accurate and reliable, timely, coherent, comparable across regions and countries, and readily accessible by users, i.e. the Principles of Statistical Output.

PRINCIPLE 11

Relevance

INDICATOR European Statistics meet the needs of users.

11.1 Procedures are in place to consult users, to monitor the relevance and value of existing statistics in meeting their needs, and to consider and anticipate their emerging needs and priorities. Innovation is pursued to continuously improve statistical output.

11.2 Priority needs are being met and reflected in the work programme.

11.3 User satisfaction is monitored on a regular basis and is systematically followed up.

PRINCIPLE 12

Accuracy and Reliability

INDICATOR European Statistics accurately and reliably portray reality.

12.1 Source data, integrated data, intermediate results and statistical outputs are regularly assessed and validated.

12.2 Sampling errors and non-sampling errors are measured and systematically documented according to the European standards.

12.3 Revisions are regularly analysed in order to improve source data, statistical processes and outputs.

PRINCIPLE 13

Timeliness and Punctuality

INDICATOR European Statistics are released in a timely and punctual manner.

13.1 Timeliness meets European and other international release standards.

13.2 A standard daily time for the release of statistics is made public.

13.3 The periodicity of statistics takes into account user requirements as much as possible.

13.4 Divergence from the dissemination time schedule is publicised in advance, explained and a new release date set.

13.5 Preliminary results of acceptable aggregate accuracy and reliability can be released when considered useful.

PRINCIPLE 14

Coherence and Comparability

INDICATOR European Statistics are consistent internally, over time and comparable between regions and countries; it is possible to combine and make joint use of related data from different data sources.

14.1 Statistics are internally coherent and consistent (i.e. arithmetic and accounting identities observed).

14.2 Statistics are comparable over a reasonable period of time.

14.3 Statistics are compiled on the basis of common standards with respect to scope, definitions, units and classifications in the different surveys and data sources.

14.4 Statistics from the different data sources and of different periodicity are compared and reconciled.

14.5 Cross-national comparability of the data is ensured within the European Statistical System through periodical exchanges between the European Statistical System and other statistical systems. Methodological studies are carried out in close co-operation between the Member States and Eurostat.

PRINCIPLE 15

Accessibility and Clarity

European Statistics are presented in a clear and understandable form, released in a suitable and convenient manner, available and accessible on an impartial basis with supporting metadata and guidance.

INDICATOR	
15.1	Statistics and the corresponding metadata are presented, and archived, in a form that facilitates proper interpretation and meaningful comparisons.
15.2	Dissemination services use modern information and communication technology, methods, platforms and open data standards.
15.3	Custom-designed analyses are provided when feasible and the public is informed.
15.4	Access to microdata is allowed for research purposes and is subject to specific rules or protocols.
15.5	Metadata related to outputs are managed and disseminated by the statistical authority according to the European standards.
15.6	Users are kept informed about the methodology of statistical processes including the use and integration of administrative and other data.
15.7	Users are kept informed about the quality of statistical outputs with respect to the quality criteria for European Statistics.

For more information on the European Statistics Code of Practice
http://ec.europa.eu/eurostat/web/quality/overview
Email: ESTAT-QUALITY@ec.europa.eu



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